

Exploiting Separability in Multi-Scale Grey-Box Bayesian Optimization

Joshua E. Hammond¹
Brian A. Korgel^{1,3}

Tyler A. Soderstrom²
Michael Baldea^{1,4*}

¹McKetta Department of Chemical Engineering, The University of Texas at Austin, 200 E. Dean Keaton St. Stop C0400, Austin, Texas 78712, USA

²ExxonMobil Technology and Engineering, 22777 Springwoods Village Pkwy, Spring, Texas 77389, USA

³Energy Institute, The University of Texas at Austin, 2304 Whitis Ave. Stop C2400, Austin, Texas 78712, USA

⁴Institute for Computational Engineering and Sciences, The University of Texas at Austin, 201 E. 24th Street, POB 4.102, Stop C0200, Austin, Texas 78712, USA

Abstract

We consider grey-box optimization problems where the decision variables naturally partition into black-box variables x^{BB} (~~entering an expensive~~ as arguments to an expensive black-box function) and white-box variables x^{WB} (~~governed by known equations, with the~~ governed by a set of explicit, closed-form equations that also depend on the output of the black-box ~~depending only on x^{BB} function~~). We exploit this separability through a bilevel reformulation: ~~the outer loop uses an outer~~ Bayesian optimization (BO) to optimize the scalar objective as a function of x^{BB} alone, while ~~the inner loop implicitly an inner problem~~ solves the white-box subproblem via ~~nonlinear programming (NLP)~~ global optimization. The Gaussian process surrogate ~~therefore operates over used in BO~~ is therefore defined $\mathbb{R}^{n_{\text{BB}}}$ rather than $\mathbb{R}^{n_{\text{WB}}+n_{\text{BB}}}$, and white-box constraints are satisfied exactly whenever the inner ~~NLP optimizer~~ converges to a feasible point—without penalty functions, chance constraints, or moment approximations. On a suite of 13 benchmark problems, bilevel BO achieves ~~14×–277~~ lower regret, ~~000× lower regret than full-space BO across 7,410 independent optimization runs, with comparable wall time for 12 of 13 problems. The advantage grows with dimension—up to 277,000× on 5D problems—and~~ with fewer iterations and wall clock time. This advantage is robust to initialization ~~size and exploration~~ parameters. When grey-box problems exhibit variable separability, solving the known subproblem exactly rather than surrogating it should be ~~the default approach~~ set size, exploration parameters, and inner-solver choice.

Keywords: Bayesian optimization, Grey-box optimization, Bilevel optimization, Surrogate-based optimization, Multi-scale design, Dimensionality reduction

*Corresponding author: mbaldea@che.utexas.edu

1 Introduction

Surrogate-based optimization is the standard approach for expensive black-box functions [1], yet many problems contain known, differentiable substructure that monolithic surrogates waste samples learning. When ~~an optimized system decomposes into an expensive~~ a model can be readily separated into an “expensive” black-box component and a ~~cheap~~ “cheap” white-box component, building a single surrogate over all variables forces the ~~Gaussian-process (GP) surrogate model~~ to learn both the unknown and the known ~~simultaneously~~ components. The cost of this redundancy grows with dimension: as the size of the combined variable space $[x^{\text{WB}}, x^{\text{BB}}]$ expands, the curse of dimensionality ~~penalizes~~ burdens the surrogate for structure it should never have had to discover.

This situation arises naturally in multi-scale engineering problems where design and operation are jointly optimized. For example, a catalyst designer uses density functional theory (DFT) simulations to predict kinetic parameters, which are then used in well-understood reactor mass and energy balances to evaluate reaction yield. Similarly, a polymer engineer uses molecular simulations to estimate membrane transport properties, then solves the solution-diffusion equations to size a separation unit. In each case, the overall objective depends on two groups of decision variables. *Black-box variables* x^{BB} are inputs to expensive simulations or experiments ~~such as molecular descriptors, catalyst properties, or experimental parameters~~. Conversely, *white-box variables* x^{WB} ~~are decision variables that interact with well-understood equations~~ are variables present in the closed-form macroscopic equation. The black-box and white-box ~~components~~ elements of the model are *separable*: the expensive computation depends only on x^{BB} , producing outputs $y = f^{\text{BB}}(x^{\text{BB}})$ that parameterize the white-box model. The key structural property is that, for any fixed x^{BB} and y , the remaining optimization over x^{WB} is a standard NLP solvable by conventional methods assuming that the WB equations are continuous and that x^{WB} are all real.

Several lines of work exploit known structure in expensive optimization, ranging from deterministic grey-box solvers [2, 3, 4] to composite and structured Bayesian optimization [5, 6, 7] to data-driven bilevel methods [8, 9]. We review these in Section 3. The common gap is that no prior method combines a ~~GP-based surrogate-based~~ global search over only the black-box variables with exact NLP solution of the white-box ~~subproblem—the subproblem—a~~ configuration that fully exploits variable separability for both dimensionality reduction and exact constraint satisfaction.

When the white-box optimization is ~~solvable exactly~~ solved explicitly, the surrogate model should operate over $\mathbb{R}^{n_{\text{BB}}}$ rather than $\mathbb{R}^{n_{\text{WB}}+n_{\text{BB}}}$. We exploit this through a bilevel reformulation: the outer loop uses ~~BO~~ Bayesian Optimization (BO) to search over x^{BB} ; for each candidate, we evaluate the black box to obtain intermediate outputs y , then solve the white-box optimization exactly via ~~nonlinear programming (NLP)~~ global optimization to get $x^{\text{WB}*}$ and ~~$J(x^{\text{WB}*})$~~ the value of the objective function. The GP models only the scalar mapping ~~$x^{\text{BB}} \mapsto J(x^{\text{WB}*}, y, x^{\text{BB}})$~~ $J(x^{\text{WB}*}, y, x^{\text{BB}}) \mapsto \mathbb{R}$, reducing the surrogate dimension from $n_{\text{WB}} + n_{\text{BB}}$ to n_{BB} and satisfying white-box constraints exactly whenever the inner ~~NLP optimizer~~ converges to a feasible point.

We make three contributions:

1. A bilevel reformulation of the class of that reduces surrogate dimension-

ality from $n_{\text{WB}} + n_{\text{BB}}$ to n_{BB} by solving the white-box subproblem via global optimization, with white-box constraints satisfied exactly whenever the inner optimizer converges to a feasible point—without penalty functions, chance constraints, or moment approximations.

2. A benchmark suite of 13 separable grey-box problems (2–5 total variables, 0–3 constraints) considering synthetic test functions and prototype engineering applications—the largest such suite for this problem class.
3. Comprehensive empirical evidence from 8,450 independent optimization runs (8,190 BO + 130 NLP + 130 BH): the proposed bilevel strategy achieves $11 \times 10^8 \times$ lower regret than black-box BO across all 13 problems, with equal or faster wall time and robustness to hyperparameters (n_{init} , ξ) and inner-solver choice.

2 Problem Formulation

We consider optimization problems in which an expensive black-box model is coupled with known, differentiable equations. This setting arises in multi-scale engineering design—molecular simulations feeding into process models—but also in any domain where part of the system is analytically tractable and part is not. We first formalize the general problem, then identify the structural property our method exploits.

2.1 Problem Setting and Notation

The decision variables comprise two groups:

- $x^{\text{WB}} \in \mathbb{R}^{n_{\text{WB}}}$: *white-box variables* governed by known, differentiable equations (e.g., temperatures, pressures, flow rates in engineering; policy parameters in control; design dimensions in structural optimization). These variables enter the white-box model f^{WB} , the objective J , and the inequality constraints g .
- $x^{\text{BB}} \in \mathbb{R}^{n_{\text{BB}}}$: *black-box variables* that parameterize an expensive, non-differentiable function f^{BB} (e.g., molecular descriptors requiring DFT, material properties from simulations, hyperparameters of a costly simulator). These variables enter f^{BB} directly and may also appear in J and g .

The black-box model

$$y = f^{\text{BB}}(x^{\text{BB}}), \quad f^{\text{BB}} : \mathbb{R}^{n_{\text{BB}}} \rightarrow \mathbb{R}^{n_y}, \quad (1)$$

maps x^{BB} to a vector of intermediate parameters $y \in \mathbb{R}^{n_y}$ that couple the black-box and white-box components. We assume f^{BB} is expensive to evaluate and that we lack closed-form expressions or derivative information for it.

The white-box model takes the implicit form

$$f^{\text{WB}}(x^{\text{WB}}, y) = 0, \quad f^{\text{WB}} : \mathbb{R}^{n_{\text{WB}}} \times \mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_{\text{WB}}}, \quad (2)$$

encoding known equations (e.g., mass and energy balances) that relate the white-box variables x^{WB} to the black-box outputs y . We assume f^{WB} is

sufficiently smooth and that the Jacobian $\partial f^{\text{WB}}/\partial x^{\text{WB}}$ is available analytically.

The inequality constraints

$$g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0, \quad g : \mathbb{R}^{n_{\text{WB}}} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{\text{BB}}} \rightarrow \mathbb{R}^{n_g}, \quad (3)$$

encode n_g design specifications, safety limits, or feasibility requirements that may depend on both variable groups and the black-box outputs.

The complete optimization problem over both variable groups is

$$\min_{x^{\text{WB}}, x^{\text{BB}}} J(x^{\text{WB}}, y, x^{\text{BB}}) \quad (4a)$$

$$\text{s.t.} \quad f^{\text{WB}}(x^{\text{WB}}, y) = 0 \quad (4b)$$

$$y = f^{\text{BB}}(x^{\text{BB}}) \quad (4c)$$

$$g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0 \quad (4d)$$

$$x^{\text{WB}} \in \mathcal{X}^{\text{WB}}, \quad x^{\text{BB}} \in \mathcal{X}^{\text{BB}} \quad (4e)$$

The objective $J(x^{\text{WB}}, y, x^{\text{BB}})$ and inequality constraints $g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0$ may depend on the white-box variables, the black-box outputs, and—in some problems—directly on the black-box variables x^{BB} .

3 Related Work

~~Our bilevel framework~~ Problem (4) sits at the intersection of surrogate-based global optimization, grey-box Bayesian optimization, bilevel programming, and constrained expensive optimization. ~~We review each thread, focusing on the structural assumptions each approach makes and where those assumptions limit applicability to separable grey-box problems.~~ organize prior work along two axes that follow directly from Assumption 1: (i) whether the method distinguishes x^{BB} from x^{WB} , and (ii) whether the white-box subsystem f^{WB} and constraints g are exploited exactly or approximated. Methods that collapse along axis (i) pay the curse of dimensionality in $n_{\text{WB}} + n_{\text{BB}}$; methods that collapse along axis (ii) trade exactness for the generality of uncertainty propagation.

3.1 Surrogate-Based Global Optimization

The use of surrogate models to guide expensive optimization traces to the Efficient Global Optimization (EGO) algorithm of Jones et al. [10], which introduced Expected Improvement (EI) as an acquisition function for Gaussian ~~process~~ Process (GP) surrogates. EGO treats the objective as a monolithic black box and builds a single GP over the full decision ~~space—an approach that remains~~ space $\mathcal{X}^{\text{WB}} \times \mathcal{X}^{\text{BB}} \subset \mathbb{R}^{n_{\text{WB}}+n_{\text{BB}}}$ which is the default in most BO implementations [11, 1].

Within the deterministic surrogate ~~tradition~~ realm, Boukouvala, Hasan, and Floudas [2] developed a methodology for constrained grey-box problems that selects surrogates from a diverse set of functional forms and globally optimizes the resulting NLP. The companion ARGONAUT framework [3] formalized this into an iterative algorithm with variable selection, bounds tightening, and constrained sampling for problems with up to 100 variables; Kieslich et

al. [12] extended the approach using Smolyak grids and polynomial surrogates. These methods share our philosophy of exploiting known equations, but ~~use deterministic surrogates that they build surrogates over the joint $(x^{\text{WB}}, x^{\text{BB}})$ space rather than restricting the surrogate to x^{BB} as Assumption 1 permits; f^{WB} is used only to generate training data or to tighten bounds on $\mathcal{X}^{\text{WB}}$, not to eliminate x^{WB} from the surrogate’s domain. Moreover, the deterministic surrogates they employ~~ lack the principled exploration–exploitation trade-off of probabilistic models and tend to overexploit early samples under tight evaluation budgets.

The trust-region filter algorithms of Eason and Biegler [4, 13] offer a complementary local approach: they embed surrogates for ~~black-box components f^{BB} within a trust region and solve the glass-box NLP at each iteration. This is architecturally close to our inner-loop NLP solve, but the trust-region framework provides white-box NLP in x^{WB} at each iteration—architecturally close to the inner subproblem in Assumption 1(ii), but with only local convergence guarantees—no guarantees in x^{BB} rather than a global exploration mechanism steers the search away from poor basins.~~

3.2 Grey-Box Bayesian Optimization

Grey-box BO methods incorporate known structure into the surrogate to reduce its modeling burden. Astudillo and Frazier [5] ~~modeled composite functions $g(h(x))$ by placing~~ introduced Bayesian Optimization of Composite Functions (BOCF), which in our notation models $J(x^{\text{WB}}, f^{\text{BB}}(x^{\text{BB}}), x^{\text{BB}})$ by training a multi-output GP on ~~the expensive inner function h~~ f^{BB} and propagating uncertainty through ~~the known outer function g~~ J via sampling; later extensions handle function networks [14] and partial evaluations [15]. ~~COBALT~~ Paulson and Lu [6] ~~extends~~ extend this to constrained grey-box problems by combining multivariate GP models with a constrained expected utility function, using sample average approximation and chance constraints to propagate GP uncertainty through known equations. González-Garay and del Rio-Chanona ~~lez and Zavala~~ [16] propose an interconnected-systems BO framework with adaptive linearization, Winz et al. [17] modify the upper confidence bound to exploit grey-box derivative information, and ~~BONSAI exploits~~ Kudva and Paulson [7] ~~exploit~~ function-network topology for robust optimization.

All of these methods ~~surrogate the intermediate black-box~~ place a surrogate on the intermediate outputs y ~~defined in Equation (1)~~ and propagate uncertainty through ~~the white-box equations~~ Equation (4b), requiring multi-output GPs and moment approximations or sampling. When ~~Assumption 1(ii) holds—i.e., the white-box subproblem is solvable exactly via NLP—as when $\dim(x^{\text{WB}})$ is moderate—surrogating to global optimality—using a surrogate model for y~~ adds modeling complexity without clear benefit. We instead ~~surrogate the optimal value $J(x^{\text{WB}*}, y, x^{\text{BB}})$ directly, which yields~~ place a scalar GP on the value function $J(x^{\text{WB}*}(x^{\text{BB}}), f^{\text{BB}}(x^{\text{BB}}), x^{\text{BB}})$ in $\mathbb{R}^{n_{\text{BB}}}$ ~~and avoids, avoiding~~ both multi-output surrogates and uncertainty propagation (Table 1).

3.3 Bilevel Optimization

Bilevel programming has a rich history in the global optimization literature. Gümüs and Floudas [18] developed the first rigorous global optimization ap-

proach for nonlinear bilevel problems using the α BB framework. Mitsos, Chachuat, and Barton [19] extended this to bilevel dynamic optimization, Kleiniati and Adjiman [20] addressed the mixed-integer case, and ~~Faisea~~ Faisca et al. [21] proposed parametric global optimization for bilevel programs via multi-parametric programming. ~~These exact methods provide optimality guarantees but require~~ In the language of Section 2, these methods treat x^{BB} as the upper-level decision and x^{WB} as the lower-level decision, but assume analytical access to both ~~levels—an~~ f^{BB} and f^{WB} —an assumption that breaks when ~~the upper level involves~~ f^{BB} is an expensive black-box simulation.

Data-driven approaches address this gap. The DOMINO framework of Beykal et al. [8] solves bilevel mixed-integer nonlinear problems by sampling the upper-level objective and solving the lower-level problem to global optimality at each sample point, using a grey-box optimization solver (ARGONAUT/NO-MAD) for the upper-level search. DOMINO is the closest non-BO predecessor to our approach: it ~~shares our architecture of exact lower-level solves with a data-driven upper-level search. The key distinction is that DOMINO solves the~~ x^{WB} subproblem to global optimality at each sampled x^{BB} , matching Assumption 1(ii), but uses deterministic surrogates ~~for the upper level, whereas we over~~ x^{BB} . We instead use a GP with a principled acquisition function that balances exploration and exploitation—an advantage when the evaluation budget is small (<250 calls) and the risk of overexploiting early samples is high.

Within the BO literature, Kieffer et al. [9] first applied BO to bilevel problems, using EI at the outer level and ~~SLSQP~~ Sequential Least Squares Programming (SLSQP) at the inner level. This nested architecture ~~is structurally identical to ours~~ matches our outer-over- x^{BB} /inner-over- x^{WB} decomposition, but Kieffer et al. did not exploit ~~grey-box structure or dimensionality reduction~~ the separability of Assumption 1 to reduce surrogate dimensionality from $n_{\text{WB}} + n_{\text{BB}}$ to n_{BB} . Recent work addresses the harder setting where *both* levels are black boxes: Ekmekcioglu et al. [22] model both levels as GPs over the joint space; Chew et al. [23] construct confidence-bound trusted sets with regret guarantees; Fu et al. [24] provide convergence analysis for BO bilevel problems where the inner loop is unconstrained ~~SGD~~ Stochastic Gradient Descent (SGD). When the inner problem is a known white-box NLP, solving it exactly is both cheaper and more accurate than learning a second surrogate.

Baldea [25] introduced the multi-scale bilevel BO framework we extend here, demonstrating it on a single ~~CSTR~~ chemical process case study. We advance that work with a 13-problem benchmark suite, with systematic hyperparameter sweeps, and detailed comparisons quantifying when and why bilevel decomposition helps.

3.4 Constraint Handling in Expensive Optimization

Standard constrained BO methods ~~model constraints as expensive black-box functions, each requiring a separate~~ treat each component of $g(x^{\text{WB}}, y, x^{\text{BB}})$ as an independent expensive black box with its own GP. Gardner et al. [26] introduced weighted expected improvement for unknown constraints; Gelbart et al. [27] extended this with probability-of-feasibility weighting; Picheny et al. [28] proposed a slack-variable augmented Lagrangian approach. These methods are appropriate when constraint functions are truly unknown, but when ~~constraints arise from known white-box~~ components of g depend on x^{WB}

and y through known equations—mass balances, safety envelopes, thermodynamic limits—surrogating them wastes evaluations limits—creating surrogates increases computational cost and introduces approximation error.

Our inner NLP satisfies global optimizer enforces the white-box constraints exactly—components of g exactly via Assumption 1(ii), avoiding penalty functions, chance constraints, and moment approximations. The advantage is largest when feasible regions are tight: on-in such cases(reflected in the Distillation and Heat-Exchanger problems in Section 6), bilevel BO achieves $>100,000\times$ lower regret than penalty-based black-box BO(Section 6).

4 Problem Formulation Method: Bilevel Bayesian Optimization

We consider optimization problems in which an expensive black-box model is coupled with known, differentiable equations. This setting arises in multi-scale engineering design—molecular simulations feeding into process models—but also in any domain where part of the system is analytically tractable and part is not. We first formalize the general problem, then identify the structural property our method exploits.

4.1 Problem Setting and Notation

The decision variables comprise two groups:

Parameters y are computed by a black-box model. This function f^{BB} is expensive to evaluate; we lack closed-form expressions or derivative information for it.

The white-box model generally takes the implicit form where $y \in \mathbb{R}^{n_y}$ is a vector of parameters that enter the white-box equations. We assume f^{WB} is sufficiently smooth and that the Jacobian $\partial f^{\text{WB}} / \partial x^{\text{WB}}$ is available analytically.

The integrated optimization problem over both variable groups is

The objective $J(x^{\text{WB}}, y, x^{\text{BB}})$ and inequality constraints $g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0$ may depend on the white-box variables, the black-box outputs, and—in some problems—directly on the black-box variables x^{BB} . The bilevel decomposition remains valid because, for any fixed x^{BB} and $y = f^{\text{BB}}(x^{\text{BB}})$, the inner optimization is over x^{WB} alone. The sets $\mathcal{X}^{\text{WB}}$ and $\mathcal{X}^{\text{BB}}$ define box constraints.

4.1 Separable Grey-Box Structure

Problem (4) possesses a structural property that we formalize as an assumption and then exploit algorithmically.

Assumption 1 (Separable grey-box structure). Problem (4) satisfies:

- (i) The black-box function f^{BB} depends only on x^{BB} . For any fixed x^{BB} and $y = f^{\text{BB}}(x^{\text{BB}})$, the resulting optimization over x^{WB} alone is a well-posed NLP.
- (ii) For any fixed x^{BB} and $y = f^{\text{BB}}(x^{\text{BB}})$, the white-box subproblem $\min_{x^{\text{WB}}} J(x^{\text{WB}}, y, x^{\text{BB}})$ subject to $f^{\text{WB}}(x^{\text{WB}}, y) = 0$, $g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0$, $x^{\text{WB}} \in \mathcal{X}^{\text{WB}}$ is solvable to global optimality (of the subproblem) by a suitable global optimizer.

Under Assumption 1, the optimization ~~decomposes~~can be decomposed as follows: for any fixed x^{BB} , the black-box evaluation $y = f^{\text{BB}}(x^{\text{BB}})$ and the inner ~~NLP optimization~~over x^{WB} are self-contained. This decoupling is the ~~structure~~structural feature we exploit: rather than optimizing all variables jointly, we can search over the low-dimensional black-box space while solving the white-box subproblem exactly at each step. Note that f^{BB} can always be augmented with identity mappings (e.g., $y_{n_y+1} = x_1^{\text{BB}}$) to absorb any direct x^{BB} dependence in J or g , so the distinction between “depends on x^{BB} through y ” and “depends on x^{BB} directly” is a modeling choice, not a structural limitation.

Despite this structure, problem (4) resists standard solution methods for two reasons. First, the black-box nature of f^{BB} precludes using gradient-based NLP solvers. Finite-difference ~~gradients~~gradient approximations require $O(n_{\text{BB}})$ evaluations of f^{BB} per iteration—prohibitive when each evaluation takes minutes to hours. Second, full-space surrogate-based optimization (EGO [10], standard BO [1]) over $[x^{\text{WB}}, x^{\text{BB}}]$ builds a GP surrogate in $\mathbb{R}^{n_{\text{WB}}+n_{\text{BB}}}$, wasting samples on the known white-box response and suffering the curse of dimensionality as ~~$n_{\text{WB}} + n_{\text{BB}}$~~ n_{WB} grows.

5 ~~Method: Bilevel Bayesian Optimization~~

4.1 Bilevel Reformulation

~~We~~Based on the assumptions in Section 4.1 we reformulate (4) as a bi-level optimization problem that separates what we know from what we must learn:

$$\min_{x^{\text{BB}}} J(x^{\text{WB}\star}, y, x^{\text{BB}}) \quad (5a)$$

$$\text{s.t. } y = f^{\text{BB}}(x^{\text{BB}}) \quad (5b)$$

$$x^{\text{WB}\star} = \arg \min_{x^{\text{WB}}} J(x^{\text{WB}}, y, x^{\text{BB}}) \quad (5c)$$

$$\text{s.t. } f^{\text{WB}}(x^{\text{WB}}, y) = 0 \quad (5d)$$

$$g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0 \quad (5e)$$

$$x^{\text{WB}} \in \mathcal{X}^{\text{WB}} \quad (5f)$$

$$x^{\text{BB}} \in \mathcal{X}^{\text{BB}} \quad (5g)$$

The **inner problem** (5c)–(5f) is a standard nonlinear program (NLP). Given fixed x^{BB} and $y = f^{\text{BB}}(x^{\text{BB}})$, it optimizes only the white-box variables x^{WB} using the analytical white box model; x^{BB} and y enter as parameters. ~~This~~Because the white-box equations are closed-form and inexpensive to evaluate, ~~this subproblem can be solved efficiently with conventional NLP solvers that exploit gradient and Hessian information to global optimality using a global optimizer such as Basin-Hopping (BH) with an SLSQP local minimizer.~~

The **outer problem** (5a)–(5b) searches over black-box decisions x^{BB} . For each candidate x^{BB} , we evaluate the black box to obtain y , solve the inner ~~NLP to problem to global optimality to~~ get $x^{\text{WB}\star}$, and record the resulting objective value $J(x^{\text{WB}\star}, y, x^{\text{BB}})$. This outer loop is where we apply Bayesian optimization.

Remark 1. The key advantage is dimensionality reduction. The BO surrogate now operates over $\mathbb{R}^{n_{\text{BB}}}$ rather than $\mathbb{R}^{n_{\text{WB}}+n_{\text{BB}}}$. ~~Since typically~~When

$n_{BB} \ll n_{WB}$, the number of black-box evaluations required to build an accurate surrogate drops substantially [relative to applying BO to problem \(4\)](#).

Remark 2. If the inner problem (5c)–(5f) is infeasible for some y , we return a large penalty value. The BO acquisition function will naturally steer away from black-box variable choices that lead to infeasible white-box solutions.

Remark 3. [The bilevel decomposition remains valid because, for any fixed \$x^{BB}\$ and \$y = f^{BB}\(x^{BB}\)\$, the inner optimization is over \$x^{WB}\$ alone. The sets \$\mathcal{X}^{WB}\$ and \$\mathcal{X}^{BB}\$ define box constraints.](#)

4.2 Algorithm, Surrogate Model, and Acquisition Function

Algorithm 1 Bilevel Bayesian Optimization

Require: Black-box bounds $\mathcal{X}^{BB}$, white-box bounds $\mathcal{X}^{WB}$, models f^{BB} , f^{WB} , J , g ; initial samples n_{init} , iterations N , exploration parameter ξ

Ensure: Best x^{BB*} , x^{WB*} , J^*

// Initialization

- 1: Sample $\{x_i^{BB}\}_{i=1}^{n_{\text{init}}}$ via Latin hypercube in $\mathcal{X}^{BB}$
 - 2: **for** $i = 1, \dots, n_{\text{init}}$ **do**
 - 3: Evaluate $y_i = f^{BB}(x_i^{BB})$
 - 4: Solve inner problem (BH): $x_i^{WB*} = \arg \min_{x^{WB}} J(x^{WB}, y_i, x_i^{BB})$ s.t.
 $f^{WB}(x^{WB}, y_i) = 0$, $g(x^{WB}, y_i, x_i^{BB}) \leq 0$
 - 5: Record $J_i = J(x_i^{WB*})$ (or penalty if infeasible)
 - 6: **end for**
 - // BO loop
 - 7: **for** $t = n_{\text{init}} + 1, \dots, n_{\text{init}} + N$ **do**
 - 8: Fit GP surrogate $\mathcal{GP}$ to $\{(x_i^{BB}, J_i)\}_{i=1}^{t-1}$
 - 9: Select $x_t^{BB} = \arg \max_{x^{BB} \in \mathcal{X}^{BB}} \alpha_{\text{EI}}(x^{BB}; \mathcal{GP}, \xi)$
 - 10: Evaluate $y_t = f^{BB}(x_t^{BB})$
 - 11: Solve inner problem (BH) $\rightarrow x_t^{WB*}$, record J_t
 - 12: Update dataset
 - 13: **end for**
 - 14: **return** best (x^{BB*}, x^{WB*}, J^*)
-

4.3 ~~Surrogate Model and Acquisition Function~~

[We describe the proposed approach in Algorithm 1 and Figure 1 and provide a discussion as follows:](#) The GP surrogate in Algorithm 1 models the scalar map-

ping $x^{BB} \mapsto J(x^{WB*}(x^{BB}), f^{BB}(x^{BB}), x^{BB}) J(x^{WB*}(x^{BB}), f^{BB}(x^{BB}), x^{BB}) : \mathbb{R}^{n_{BB}} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{WB}} \mapsto \mathbb{R}$ —

a function of x^{BB} alone. ~~We~~ [In our implementation, we](#) use a Gaussian process with an automatic relevance determination (ARD) radial basis function (RBF) kernel, where a separate length scale per dimension allows the GP to adapt to anisotropic landscapes. The outer loop selects the next evaluation by maximizing Expected Improvement (EI) with exploration parameter $\xi \geq 0$. Kernel and acquisition function details [for our implementation](#) are in Appendices C and B; here we emphasize that both are standard—the gains reported in Section 6 come from problem structure, not from novel surrogate or acquisition choices, [although we expect that adopting them would yield further benefits.](#)

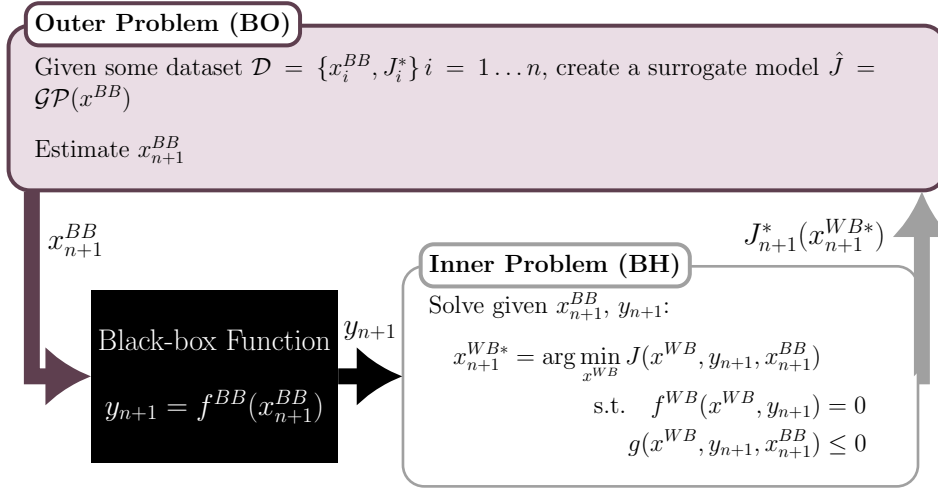


Figure 1: Flow of the multi-scale Bayesian optimization framework. The outer loop uses BO to select black-box variables x^{BB} , which are evaluated through the black-box model to produce parameters y . The inner loop solves the white-box optimization over x^{WB} via Basin-Hopping given x^{BB} and y , returning the optimal objective $J(x^{\text{WB}*}, y, x^{\text{BB}})$ to update the BO surrogate.

4.3 Key Properties

The bilevel reformulation (5) has several properties worth highlighting. The first is formalized as a proposition.

Proposition 1. *Under Assumption 1, the bilevel reformulation (5) preserves the global optimum of (4) and reduces the GP surrogate domain from $\mathbb{R}^{n_{\text{WB}} + n_{\text{BB}}}$ to $\mathbb{R}^{n_{\text{BB}}}$.*

The proof is in Appendix G. The key ~~consequence is~~ consequences are threefold:

Exact constraint satisfaction. Constraints $g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0$ are handled by the inner ~~NLP solver~~ global optimizer, not by penalty functions or chance constraints. The GP never needs to learn the constraint boundary. When the inner ~~NLP optimizer~~ converges to a globally feasible point (Assumption 1(ii)), feasibility is guaranteed; ~~in practice, local solvers such as SLSQP with multi-start may find infeasible local optima on non-convex inner problems.~~ In our implementation, Basin-Hopping (BH) [29, 30, 31, 32, 33] with Sequential Least Squares Programming (SLSQP) [34, 35, 36] local minimization provides high-confidence global solutions. We also compare a simpler solution using SLSQP with multiple restarts as a more time-efficient method to solve the reduced-dimension inner problem. Further guarantees of global optimization could be obtained with deterministic global solvers (e.g., BARON), as discussed in Section 7. Details on these solvers and implementation are available in Appendix E.

Scalar surrogate. The method requires only one GP for the scalar optimal objective $J(x^{\text{WB}*}, y, x^{\text{BB}})$. Approaches that surrogate the black-box outputs $y \in \mathbb{R}^{n_y}$ instead require a multi-output GP, with the associated challenges of cross-covariance specification and $O(n_y^3)$ scaling.

Additional implementation details—infeasibility handling and solver modularity—are discussed in Appendix ~~HE~~.

4.4 Relationship to Existing Grey-Box Methods

Table 1: Comparison of grey-box BO approaches. Bilevel BO (this work) is the only method that both separates variables and solves the white-box subproblem exactly.

	Bilevel BO (this work)	COBALT [6]	BOCF [5]	Full-space BO
What is modeled	$x^{\text{BB}} \mapsto J^*$	$x^{\text{BB}} \mapsto y$	$x \mapsto h(x)$	$x \mapsto J$
Model type	scalar GP	multi-output GP	vector GP	scalar GP
Model dimension	n_{BB}	n_{BB}	$n_{\text{WB}} + n_{\text{BB}}$	$n_{\text{WB}} + n_{\text{BB}}$
Variables separated?	Yes	Yes	No	No
Constraint handling	exact (global opt.)	chance constraints	penalty	penalty
Inner optimization	global optimization (BH)	moment propagation	none	none
y -dependent constraints	via inner optimizer	natively	penalty	penalty

Table 1 summarizes the distinctions between the proposed method and a subset of the methods reviewed in Section 3 that are most directly related and relevant. Both bilevel BO and COBALT achieve the same input dimensionality reduction to n_{BB} . The distinction is in the surrogate output (scalar vs. multi-output GP) and constraint handling (exact ~~NLP-global optimization~~ vs. moment approximation), not the input dimension. COBALT and BOCF surrogate intermediate black-box outputs y and propagate uncertainty through the white-box equations, requiring multi-output GPs and moment approximations. Our method surrogates the scalar optimal objective $J(x^{\text{WB}^*})$ directly, at the cost of an ~~NLP-inner global optimization~~ solve per outer iteration. When the white-box model is moderately sized and the black-box evaluation ~~dominates cost~~ is the primary computational expense, this overhead is negligible. The approaches are complementary: COBALT handles y -dependent constraints natively and enables risk-averse decisions via uncertainty propagation; BOCF handles non-separable composite functions. Our method trades their generality for simplicity and exactness when Assumption 1 holds.

5 Benchmark Problem Suite

~~No existing benchmark suite targets the separable grey-box structure of Assumption 1: an outer black-box mapping $y = f^{\text{BB}}(x^{\text{BB}})$ coupled with an inner white-box optimization over x^{WB} , subject to inequality constraints. Existing suites for composite BO do not separate variables; those used in grey-box BO contain only a handful of problems; and standard global optimization testbeds (CEC, COCO/BBOB) lack bilevel structure entirely.~~ We propose a suite of 13 problems spanning synthetic test functions and engineering domains, summarized in Table 2. Full mathematical formulations are in Appendix A.

5.1 ~~Problem Summary~~

5.1 ~~Design Principles~~

The suite is designed around five principles: (1) every problem has a verified global optimum (Appendix F); (2) constraints are physically motivated;

(3) black-box functions encode realistic structure-property relationships (Sabatier volcano curves, Langmuir isotherms, Robeson upper bounds, Hansen solubility); (4) problems span a range of difficulty (unconstrained to 3 constraints, 2D to 5D, min and max); and (5) all problems are cheap to evaluate, enabling the ~~108,920-run~~ 450-run statistical comparison in Section 6.

To our knowledge, no existing benchmark suite targets a separable grey-box structure that reflects Assumption 1: an outer black-box mapping $y = f^{\text{BB}}(x^{\text{BB}})$ coupled with an inner white-box optimization over x^{WB} , subject to inequality constraints. Existing problem suites for composite BO [5] do not separate variables; those used in grey-box BO [6] contain only a handful of problems; and standard global optimization testbeds such as the IEEE Congress on Evolutionary Computation (CEC) benchmarks [37] and the Comparing Continuous Optimizers (COCO) platform with its Black-Box Optimization Benchmarking (BBOB) suite [38] lack bilevel structure entirely.

Table 2: Summary of the 13-problem benchmark suite. n_{WB} : white-box variables, n_{BB} : black-box variables, n_y : black-box outputs, n_g : inequality constraints.

Problem	n_{WB}	n_{BB}	n_y	n_g	Type	Domain
Small-Feasible-Region 1	1	1	1	1	min	Synthetic
Small-Feasible-Region 2	1	1	1	1	min	Synthetic
Rastrigin	2	1	1	0	min	Multimodal
Toy-Hydrology	1	1	1	2	min	Hydrology
Rosen-Suzuki	2	2	2	3	min	Constrained
CSTR	3	2	3	2	max	Catalysis
Heat-Exchanger	3	2	3	2	min	Heat Exchanger Network
PSA	3	2	3	2	min	Adsorption
Batch-Reactor	3	2	3	2	min	Kinetics
Distillation	3	2	3	3	min	Separation
Evaporator	3	2	3	2	min	Evaporation
Membrane	3	2	3	2	min	Membrane sep.
Williams-Otto	3	2	2	2	max	Process opt.

5.1 Representative Problems

We highlight two problems; full formulations for all 13 appear in Appendix A. Adapted from Gardner et al. [26] and Ariaifar et al. [39], these 2D ~~problems minimize the constraints of these problems create~~ constrained minimization problems feature small, disconnected feasible regions. The two-dimensional structure permits direct visualization of the search behavior (Section 6.2).

6 Computational Experiments

We evaluate bilevel BO against full-space baselines on the 13-problem benchmark suite. Section 6.1 specifies the solvers, metrics, and hyperparameter sweep. Section 6.2 then uses the two Small-Feasible-Region problems to visualize how bilevel and black-box BO explore the search space differently. Section 6.3

reports final regret across all 13 benchmarks, convergence speed, and robustness to hyperparameter and dimension.

6.1 Experimental Design

This section specifies the four solvers compared (Section 6.1.1) and the four metrics reported (Section 6.1.2). Per-problem hyperparameter settings and full sweep results are deferred to Appendix J.

6.1.1 Solvers Compared

We compare ~~three~~ four solvers that represent different ways of handling the grey-box structure in problem (4).

The first solver, **black-box NLPBH**, ~~treats f^{BB} as a numerically differentiable function and applies multi-start SLSQP~~ applies SciPy’s Basin-Hopping (BH) solver over the full variable space $[x^{\text{WB}}, x^{\text{BB}}]$ with an SLSQP local minimizer. Basin-Hopping performs a global search by alternating random perturbations (uniform steps clipped to variable bounds) with local SLSQP minimizations, accepting or rejecting each local minimum via a Metropolis criterion ($T = 100$). ~~We use 20 random restarts drawn from a Latin hypercube design.~~ This solver provides an “oracle” baseline: it has gradient access (via finite differences) to both the white-box and black-box models. In practice, finite-differencing the black box requires $O(n_{\text{BB}})$ evaluations per gradient step, so this approach is ~~expensive when a global search baseline~~. Basin-Hopping is gradient-free at the global level and explores the full feasible region. However, it evaluates f^{BB} ~~is~~ costly at every local minimization, so the total number of black-box evaluations grows with the number of Basin-Hopping iterations.

The second solver, **black-box BO (BB-BO)**, applies standard Bayesian optimization over the full $(n_{\text{WB}} + n_{\text{BB}})$ -dimensional space. A single GP surrogate models the mapping $[x^{\text{WB}}, x^{\text{BB}}] \mapsto J$. Constraints are handled via a penalty function. This solver ignores the separable structure entirely and serves as the BO baseline.

The third solver, **bilevel BO with multi-start SLSQP (Bi-BO (SLSQP))**, is the proposed method with a gradient-based inner solver. It places the GP surrogate over only the n_{BB} -dimensional black-box space and solves the constrained white-box subproblem ~~exactly via NLP at each iteration, using 50 Latin hypercube restarts of SLSQP, returning the best feasible solution found,~~ as described in Algorithm 1.

~~All three~~ The fourth solver, **bilevel BO with Basin-Hopping (Bi-BO (BH))**, is identical to the third except that it uses Basin-Hopping (100 iterations, SLSQP local minimizer, Metropolis temperature $T = 100$) as the inner-loop global solver. Both bilevel variants share the same outer-loop GP surrogate and EI acquisition function; they differ only in the inner-loop solver used for the white-box subproblem.

All four solvers use Expected Improvement (EI) as the acquisition function for the BO variants. ~~Both~~ All three BO solvers share the same ARD RBF kernel and GP configuration (Appendix C). We sweep over initial sample sizes $n_{\text{init}} \in \{1, 5, 20, 50\}$ ~~$n_{\text{init}} \in \{5, 25, 50\}$~~ and exploration parameters $\xi \in \{0.001, \dots, 1.0\}$, yielding ~~78,410~~ 450 independent runs across all 13 problems (8,190 BO + 130 NLP + 130 BH; Appendix J).

6.1.2 Metrics

We report four metrics. *Simple regret* $r_t = |J_{\text{best}}(t) - J^*|$ measures the gap between the best objective found after t evaluations and the verified global optimum J^* . *Convergence speed* is the number of iterations needed to reduce regret to 1% of its initial value. *Wall time* is the total elapsed time for $n_{\text{init}} + 200$ evaluations. Finally, we count the total number of *black-box evaluations* (calls to f^{BB}) to assess sample efficiency.

6.2 Mechanism Visualization

We begin with the two Small-Feasible-Region problems because their two-dimensional structure permits direct visualization of how bilevel BO searches differently from black-box BO. Both problems have one white-box variable (x^{WB}) and one black-box variable (x^{BB}), with small, disconnected feasible regions defined by nonlinear constraints.

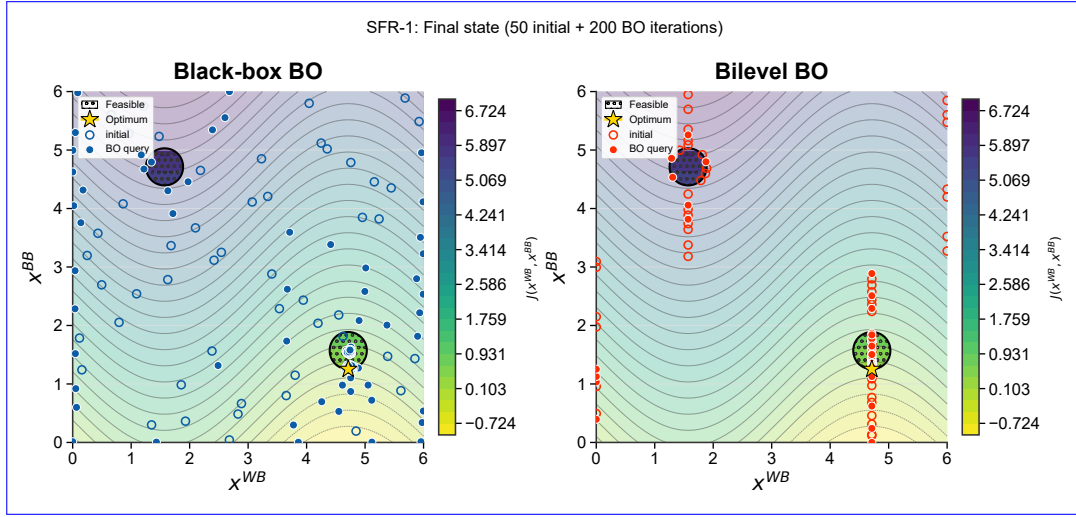


Figure 2: Small-Feasible-Region 1: search behavior of black-box BO (left) and bilevel BO (right) after 50 initial samples + 200 BO iterations. Background: objective $J(x^{\text{WB}}, y)$ (yellow = low); white regions are infeasible; hatching marks feasible islands; $\star$ marks the global optimum. Black-box BO scatters queries across the full $(x^{\text{WB}}, x^{\text{BB}})$ space. Bilevel BO operates along the x^{BB} axis—each query selects an x^{BB} value, and the inner BH solver finds the best feasible x^{WB} .

Figure 2 shows the key difference. In SFR-1, the black-box function is the identity ($y = x^{\text{BB}}$), so the bilevel decomposition reduces to: given a candidate y -value from the outer loop, find the x^{WB} that minimizes $\sin(x^{\text{WB}}) + y$ within the feasible set. The inner NLP-BH solver handles constraint satisfaction exactly, so every bilevel BO query lands in a feasible region. Black-box BO, by contrast, must jointly learn the objective function landscape *and* the constraint boundary in two dimensions, wasting many samples in infeasible regions.

SFR-2 (Figure 3) introduces a nonlinear black-box function, making the mapping from x^{BB} to the feasible objective landscape more complex. The same mechanism applies: the outer loop proposes x^{BB} , the black box computes function is used to compute y , and the inner NLP-BH solver finds the best

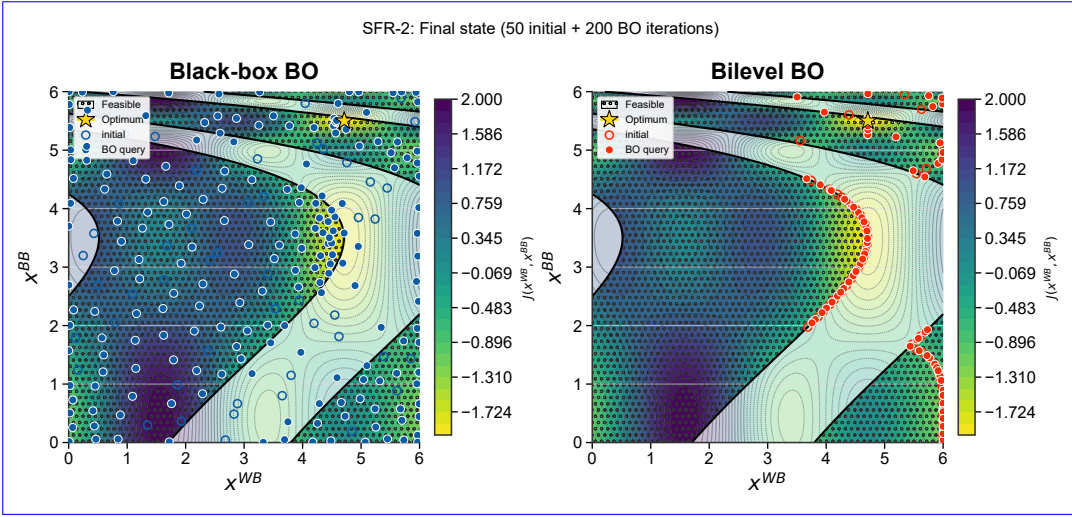


Figure 3: Small-Feasible-Region 2: same layout as Figure 2. Here the black-box function is nonlinear ($y = \frac{x^{\text{BB}} - 5.5}{2} \exp(x^{\text{BB}}/2)$), distorting the relationship between x^{BB} and the feasible set. Bilevel BO still operates near the optimum because the GP surrogate learns the scalar mapping $x^{\text{BB}} \mapsto J(x^{\text{WB}*})$ in one dimension.

feasible x^{WB} . The GP surrogate now models a one-dimensional function $x^{\text{BB}} \mapsto J(x^{\text{WB}*})$ rather than the two-dimensional objective-plus-constraint landscape that black-box BO must learn.

The convergence curves (Figure 4) confirm the visual impression. On SFR-1, ~~bilevel-BO reduces regret to $\sim 10^{-1.5}$ while Bi-BO (SLSQP) reduces regret by an order of magnitude relative to black-box BO plateaus near $10^{0.4}$ —a 76BO—an $11\times$ improvement.~~ On SFR-2, ~~bilevel-BO Bi-BO (SLSQP) reaches the global optimum (regret $< 10^{-4} < 10^{-5}$) while black-box BO stalls at $10^{-0.9}$ near $10^{-1.6}$ —a 427,743872 $\times$ improvement.~~ In both cases, the mechanism is the same: bilevel BO needs only to learn a one-dimensional function, while black-box BO must simultaneously model a two-dimensional objective and discover the small feasible islands.

6.3 Main Results

The pattern observed on the SFR problems extends to all 13 benchmarks. Table 3 reports final regret after 200 BO iterations; Figure 5 shows convergence curves for the remaining 11 problems.

~~Bilevel-BO achieves $14\times$ – $277,000\times$ —~~

~~Both bilevel variants achieve lower regret than black-box BO across. Bi-BO (SLSQP) wins on all 13 problems. The gap opens early and widens: bilevel BO separates from black-box BO within 20–50 iterations on most problems. The largest gains (Heat-Exchanger: $277(11\times 10^8\times$ lower; geometric mean $3,000192\times$; Distillation: 173) at wall time comparable to BB-BO. Bi-BO (BH) wins on 12 of 13 (geometric mean $4,000840\times$) occur where tight constraints punish the penalty-based approach—the same mechanism illustrated by the SFR figures, but now, but Basin-Hopping’s random perturbations overshoot the narrow feasible region of SFR-1 and leave it worse than BB-BO there. Gains scale with constraint tightness: Heat-Exchanger ($10^8\times$) and Distillation~~

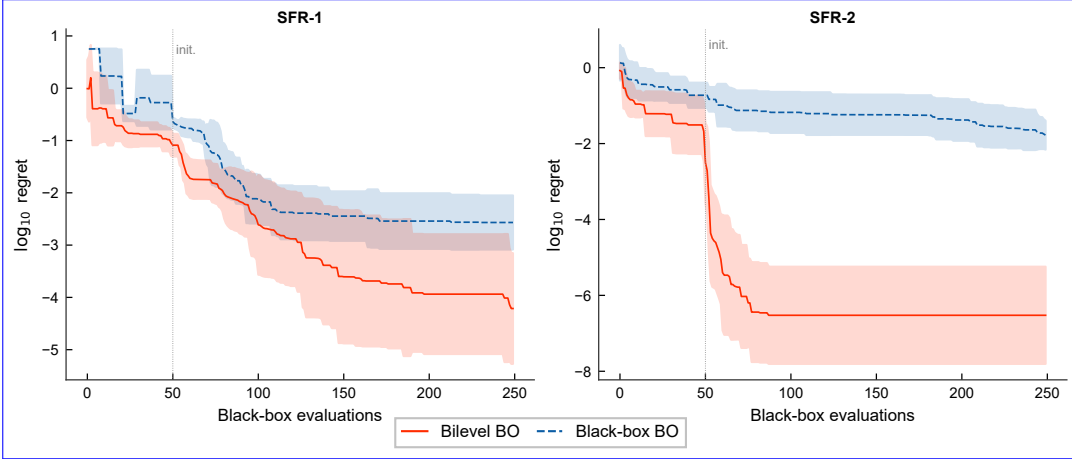


Figure 4: Regret convergence for the two Small-Feasible-Region problems. Each curve uses $n_{\text{init}} = 50$ initial Latin hypercube samples and the EI exploration parameter ξ that yielded the lowest final regret. The dashed line marks the end of initialization. Solid lines show the mean over 10 independent repetitions; shaded bands show ± 1 standard deviation. Bilevel BO (red) converges 1–2 orders of magnitude below black-box BO (blue) on both problems.

Table 3: Mean final regret and mean wall time (seconds) after 200 iterations ($n_{\text{init}} = 50$, best ξ per method, 10 repetitions; see Appendix J for per-problem ξ values). Bi-BO (SLSQP) achieves lower regret on all 13 problems; Bi-BO (BH) on 12 of 13 ([†]BH inner solver is worse than BB-BO on SFR-1).

Problem	dim	BB-BO		Bi-BO (SLSQP)		Bi-BO (BH)	
		Regret	Time (s)	Regret	Time (s)	Regret	Time (s)
Small-Feasible-Region	2	0.0048	207	0.0004	141	0.0334 [†]	826
Small-Feasible-Region-2	2	0.0243	197	0.0000	122	0.0000	150
Rastrigin	3	1.8464	246	0.0026	104	0.0000	168
Toy-Hydrology	2	0.0114	216	0.0000	116	0.0000	173
Rosen-Suzuki	4	4.8785	281	0.0243	233	0.0567	356
CSTR	5	0.3381	326	0.0019	201	0.0016	265
Heat-Exchanger	5	37.1276	284	0.0000	200	0.0000	230
PSA	5	0.1044	315	0.0030	184	0.0010	275
Batch-Reactor	5	0.2894	301	0.0005	216	0.0003	339
Distillation	5	3621.95	260	0.0034	182	0.0078	231
Evaporator	5	0.3841	307	0.0145	212	0.0254	249
Membrane	5	1.2410	274	0.0000	214	0.0000	274
Williams-Otto	5	0.8251	273	0.0000	326	0.0000	3,130

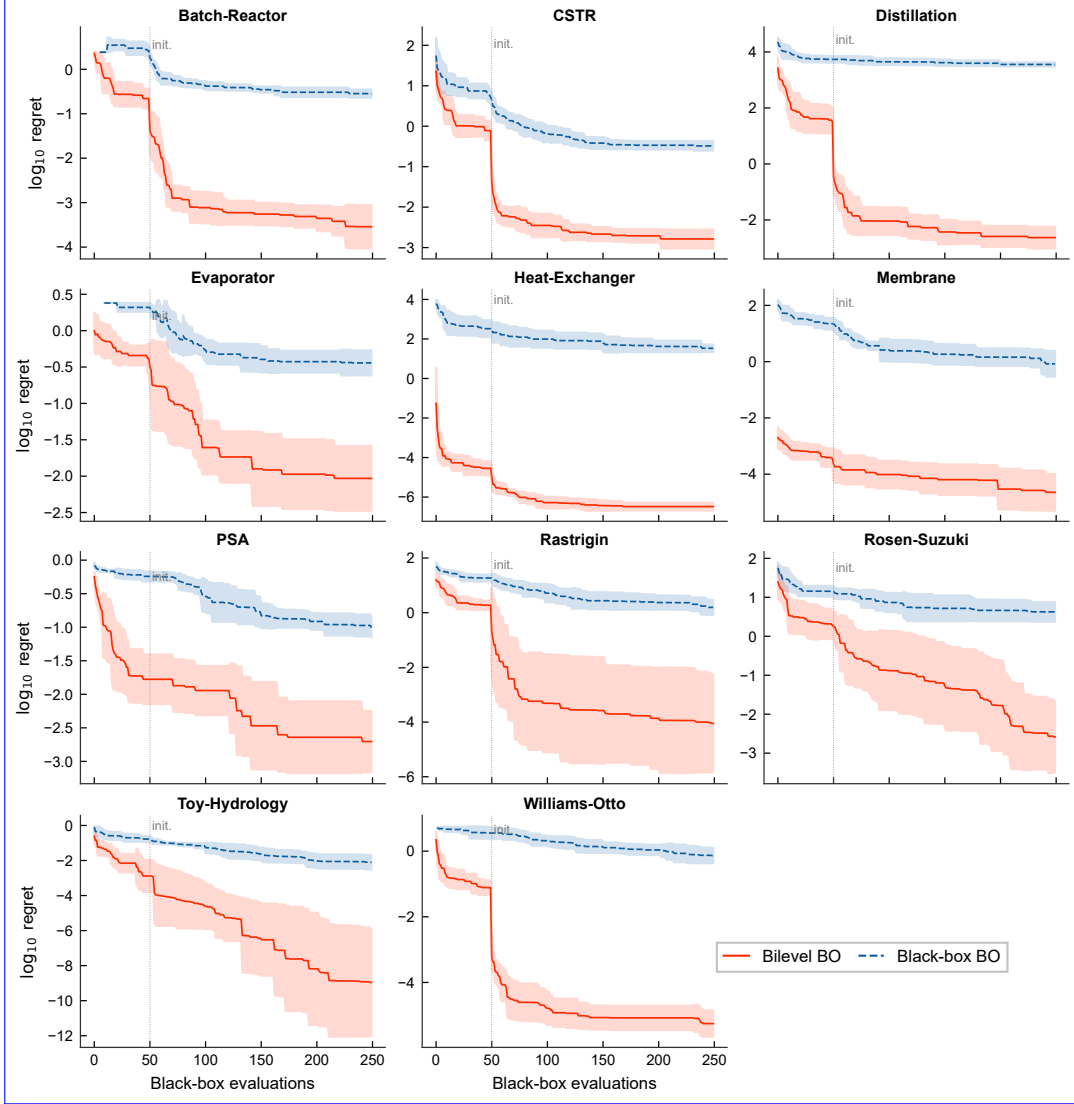


Figure 5: Regret convergence for the 11 benchmark problems not shown in Figure 4. Bi-BO (SLSQP) (red) converges to values that are 1–8 orders of magnitude below black-box BO (blue) on every problem. Each curve uses $n_{\text{init}} = 50$ initial Latin hypercube samples and the EI exploration parameter ξ that yielded the lowest final regret for that problem (see Appendix J for per-problem values). Solid lines show the mean over 10 independent repetitions; shaded bands show ± 1 standard deviation.

($10^6 \times$) expose the penalty method’s failure in five dimensions where the advantage of exact constraint satisfaction compounds with dimensionality reduction. The smallest gains (Rosen-Suzuki: 14, while the smallest SLSQP gains (SFR-1: $11 \times$, Evaporator: $1926 \times$) occur where the inner NLP has white-box subproblem itself has a narrow feasible region or multiple local optima or the constraint set is loose.

The NLP baseline fails on two problems: Rastrigin, where multimodality traps gradient-based solvers, and Small-Feasible-Region-2, where the narrow feasible region defeats finite-difference gradients. Bilevel BO Wall-time differences concentrate on problems with large infeasible regions in x^{BB} . When a sampled x^{BB} admits no feasible process variables, Basin-Hopping keeps perturbing in search of a feasible point rather than returning quickly, inflating runtime to 826 s on SFR-1 and 3,130 s on Williams-Otto (4–10 $\times$ the SLSQP inner solver). SLSQP exits on infeasibility and lets the outer BO loop advance.

The full-space BH baseline fails on Rastrigin (multimodal joint space) and performs poorly on Distillation; Bi-BO matches or approaches NLP its quality on the remaining 11 problems using ~ 250 black-box evaluations versus 900–3,000 for the 20-restart NLP the substantially higher counts BH requires. All 13 pairwise comparisons (bilevel vs. black-box BO) SLSQP pairwise comparisons are statistically significant ($p < 0.01$, Wilcoxon signed-rank test, 10 repetitions) Holm–Bonferroni adjusted $p < 0.05$; sign test 13/13, $p = 0.000244$); 12 of 13 hold for BH (sign test, $p = 0.0034$). A Friedman test across the four methods confirms differences ($\chi^2 = 17.03$, $p = 0.0007$), with Nemeyi post-hoc placing BB-BO below both Bi-BO variants and NLP (critical difference = 1.30).

6.4 Convergence Speed

Where both methods reach 1% of initial regret, bilevel BO converges $1.5 \times$ – $22 \times$ faster (Appendix Table 19). The most striking case is Williams-Otto, where bilevel BO converges in 9 iterations (~ 14 total evaluations) while black-box BO never reaches the target. Several problems do not reach the 1% threshold within 200 iterations for either method, yet bilevel BO still achieves orders-of-magnitude lower final regret (Table 3).

6.4 Scaling and Robustness

6.3.1 Dimensionality Scaling

The upward trend in Figure 6.3.1 is expected—bilevel BO searches $n_{\text{BB}} = 1$ –2 dimensions regardless of total problem size—but the spread at 5D is instructive: constraint activity and landscape ruggedness cause the ratio to vary from $19 \times$ (Evaporator) to $277,000 \times$ (Heat-Exchanger) at the same total dimension. Figure 6 confirms that this dominance holds across all hyperparameter settings: every (problem, n_{init} , ξ) configuration lies below the diagonal.

6.3.1 Computational Cost

Both BO methods require ~ 20 – 30 s wall time for 200 iterations on 12 of 13 problems, as the lower-dimensional GP in bilevel BO roughly offsets the inner NLP cost. The exception is When using BH as the inner solver for Bi-BO, the performance on the small feasible region problem (SFR-1) is worse

than the black-box BO baseline. Similarly, the solution times for the small feasible region problem and the Williams-Otto (111s vs. 26s), where the inner NLP solves a 6-species steady-state mass balance at each iteration. Both BO methods use ~ 205 problem evaluations, which are significantly higher for the Bi-BO (BH) variant compared to both the Bi-BO (SLSQP) variant and the black-box evaluations—an order of magnitude fewer than the 900–3,000 required by the multi-start NLP baseline BO baseline. Both of these problems are characterized by narrow feasible regions, and regions which are completely infeasible for certain x^{BB} values. The random perturbations in the BH algorithm cause it to overshoot these narrow feasible regions, or to iterate until satisfied that no feasible solution exists. This highlights a key tradeoff between the two inner solvers: BH provides stronger global optimization guarantees but can be inefficient in problems with narrow or disconnected feasible regions, while SLSQP is more efficient but may struggle with multimodality. The choice of inner solver should therefore be informed by problem characteristics.

6.3.1 Hyperparameter Robustness

Figure 6.3.1 shows that bilevel BO performance is nearly flat across $n_{\text{init}} \in \{1, 5, 20, 50\}$ (numerical values in Appendix Table 21). Even $n_{\text{init}} = 1$ —a single initial sample—performs comparably to $n_{\text{init}} = 50$, so fewer expensive initial evaluations suffice when f^{BB} is costly.

Bilevel BO also dominates black-box BO across all values of the EI exploration parameter. Figure 6 visualizes the relationship between final regret for black-box BO and Bi-BO (SLSQP) across all 273 (problem, n_{init} , ξ tested. No single ξ is universally optimal—the best setting ranges from $\xi = 0.001$ (CSTR, Distillation, Williams-Otto) to $\xi = 0.5$ (Small-Feasible-Region-1, Membrane)—but adaptive ξ strategies are not critical for the bilevel approach (Appendix Figure 9)—configurations. Every point lies below the diagonal, confirming that Bi-BO outperforms BB-BO regardless of hyperparameter choice. The wide range of regret ratios ($11 \times$ – $10^8 \times$) is visible in the vertical spread of points, with the largest gains occurring on problems with tight constraints and high dimensionality.

7 Discussion

Bilevel BO achieves 14× lower regret than black-box BO on every problem tested, with comparable computational cost (SLSQP variant; geometric mean 3.192×), with equal or faster wall time on 12 of 13 problems. The source of this advantage is structural, not algorithmic: we use the same GP kernel, the same acquisition function, and the same optimizer as the black-box baseline. The gains come entirely from exploiting the separable problem structure—reducing the surrogate dimension and satisfying the white-box constraints exactly. Three additional analyses—feasibility rates, a two-factor decomposition of the advantage, the relationship between regret ratio and problem characteristics, and sample efficiency crossover—provide mechanistic insight into when and why the bilevel decomposition helps.

Constraint satisfaction and wasted evaluations. A key mechanism underlying the regret gap is that black-box BO wastes evaluations on infeasible queries. Figure 7 reports the fraction of all 250 evaluations that land in feasible regions, reconstructed from stored search histories. Black-box BO wastes

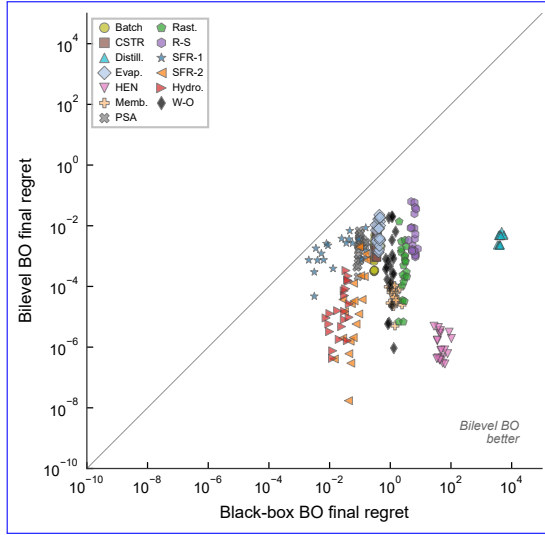


Figure 6: Scatter plot of final regret: black-box BO vs. bilevel BO (SLSQP variant). Each point is one (problem, n_{init} , ξ) configuration across 273 total configurations. All points lie below the diagonal, confirming that Bi-BO (SLSQP) outperforms BB-BO regardless of hyperparameter choice.

6–72% of its budget on infeasible points depending on the problem, with Small-Feasible-Region (72%), Batch-Reactor (71%), and Evaporator (58%) being the worst cases. Bilevel BO achieves 100% feasibility on 8 of 12 constrained problems because the inner NLP-BH solver handles constraint satisfaction exactly. On the remaining four problems (Batch-Reactor, Evaporator, Small-Feasible-Region, PSA), bilevel BO feasibility ranges from 59–91%; ~~reflecting cases where the inner NLP converges to boundary solutions with slight numerical infeasibility or where some~~. This occurs for two reasons: (1) the inner BH may return solutions where the total constraint violation $\max_i g_i(x^{\text{WB}}, y)$ is positive but small (on the order of 10^{-4} – 10^{-2}), placing the solution near but outside the feasible boundary; or (2) certain $y = f^{\text{BB}}(x^{\text{BB}})$ values ~~render the inner problem infeasible entirely~~ yield an inner problem with no feasible solution, because no $x^{\text{WB}} \in \mathcal{X}^{\text{WB}}$ satisfies $g(x^{\text{WB}}, y) \leq 0$. This is not a limitation of the bilevel architecture itself but of the ~~local NLP inner~~ solver; deterministic global ~~inner solvers~~ solvers (e.g., BARON) would restore the 100% guarantee. **Disentangling dimensionality reduction from constraint satisfaction factors influencing the regret ratio.** The regret ratios in Table 3 vary by four orders of magnitude across problems of the same total dimension. Figure 7 decomposes this variation by plotting total dimension against the feasible fraction of the domain (estimated by 10^5 uniform random samples), with the regret ratio encoded as color. Two patterns emerge. First, dimensionality reduction is the dominant factor: Factors beyond raw problem dimensionality such as constraint geometry, inner-problem multimodality, and black-box landscape complexity determine the magnitude of improvement from BB-BO to Bi-BO. For example, Rastrigin (unconstrained, 100% feasible domain) achieves a $57,7173\times$ improvement (at $n_{\text{init}} = 50$, $\xi = 0.2$; with the SLSQP inner solver (Table 3) purely from reducing the GP surrogate from 3D to 1D. Second, constraint tightness modulates but does not determine the gain: Evaporator and Batch-Reactor have the *tightest* constraints ($<1.1\%$ feasible domain) yet the *smallest* regret ratios (6. Meanwhile, Evaporator and

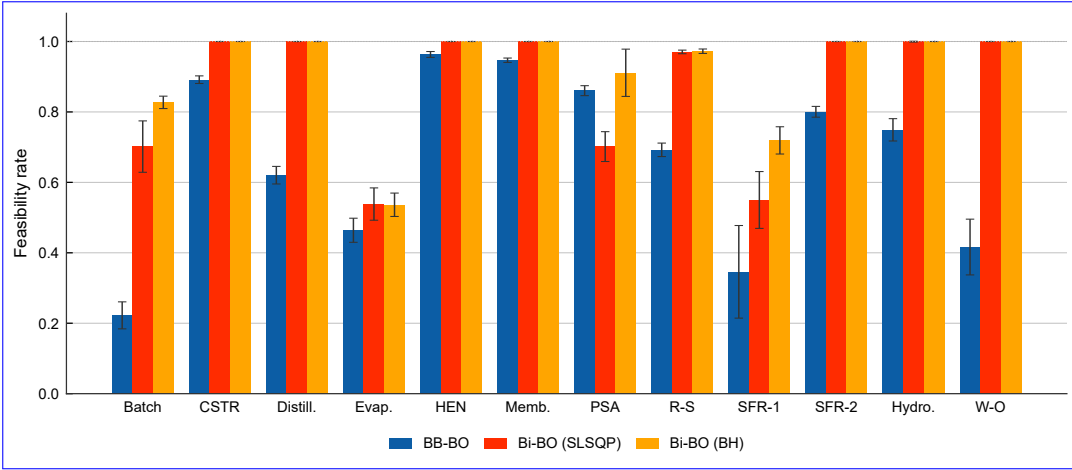


Figure 7: Fraction of evaluations landing in feasible regions across 12 constrained problems ($n_{\text{init}} = 50$, best ξ , mean over 10 repetitions). Black-box BO (blue) wastes 6–72% of its budget on infeasible queries; bilevel BO (red and orange) achieves $\geq 59\%$ feasibility on all problems and 100% on 8 of 12.

Rosen-Suzuki have the smallest SLSQP ratios ($26\times$ and $41201\times$, respectively), because their non-convex inner landscapes cause the NLP-multi-start SLSQP to find local optima. Conversely, Heat-Exchanger has a loose constraint set (93% feasible) but the largest gain ($277,000\times 10^8\times$), driven entirely by dimensionality reduction from 5D to 2D combined with a well-behaved inner landscape. The analysis confirms that when the inner NLP is well-behaved optimizer converges reliably, dimensionality reduction alone produces large gains; exact constraint satisfaction provides additional benefit but is neither necessary nor sufficient.

Sample efficiency. Beyond final regret, practitioners care about how quickly bilevel BO are interested in how quickly a method reaches useful solutions. Figure 8 reports the number of evaluations at which bilevel BO first matches the final regret that black-box BO achieves after all 250 evaluations. On 5 of 13 problems (Heat-Exchanger, Membrane, Williams-Otto, Distillation, CSTR), bilevel BO matches or surpasses black-box BO’s final performance within the initial sampling phase alone—effectively from the first BO iteration. Even in the hardest case (Evaporator), crossover occurs at 47 evaluations (19% of the total budget). This has direct practical implications: when each black-box evaluation costs hours of compute time, bilevel BO can deliver black-box BO-quality solutions using 80–100% fewer expensive evaluations.

Connection to decomposition in global optimization. As discussed in Section 3, the bilevel reformulation belongs to a broader family of decomposition strategies—block coordinate descent [40], trust-region filter methods [4, 13], and data-driven bilevel solvers like DOMINO [8]. Our results provide empirical support for a principle from bilevel optimization theory [41]: solving the inner problem exactly improves outer-level convergence. In the surrogate-based setting, the inner NLP-global optimizer eliminates noise from outer-level observations, giving the GP a cleaner signal. This suggests that bilevel decomposition may be beneficial whenever part of the optimization landscape admits an efficient exact solver, even beyond the grey-box problems studied here.

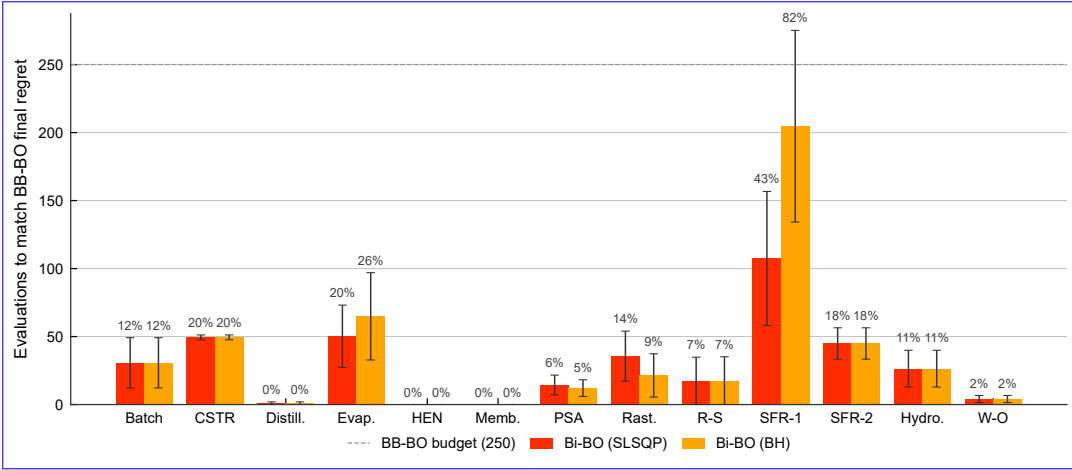


Figure 8: Sample efficiency crossover: number of evaluations for bilevel BO to match black-box BO’s final regret (at 250 evaluations). Percentages show the fraction of the total budget required. On 10 of 13 problems, bilevel BO matches black-box BO’s final performance within 10% of the evaluation budget.

Inner-solver choice: SLSQP vs. Basin-Hopping. The choice of inner-loop solver affects both solution quality and computational cost. Across all 13 problems, multi-start SLSQP and Basin-Hopping achieve comparable final regret (family-level sign test: SLSQP wins 6, BH wins 7, $p = 1.0$; geometric mean SLSQP/BH ratio: $1.52\times$). However, the two solvers exhibit complementary strengths. BH excels on multimodal inner landscapes: on Rastrigin, Bi-BO (BH) achieves $7.2 \times 10^7\times$ improvement over BB-BO, while Bi-BO (SLSQP) achieves $698\times$ —a gap of five orders of magnitude attributable entirely to SLSQP’s inability to escape local optima in the multimodal Rastrigin inner problem. Conversely, SLSQP excels on problems with narrow feasible regions: on SFR-1, Bi-BO (SLSQP) achieves $11\times$ improvement while Bi-BO (BH) is $0.1\times$ (*worse* than BB-BO), because Basin-Hopping’s random perturbations overshoot the small feasible set. SLSQP is also faster on all 13 problems (geometric mean $1.74\times$), making it the recommended default. BH is preferred when the inner landscape is known or suspected to be multimodal.

Complementarity with existing grey-box BO. As detailed in Section 3, COBALT [6] and BOCF [5] address settings our method does not: non-separable composite functions and risk-averse decisions via uncertainty propagation. Our method trades their generality for simplicity and exactness when Assumption 1 holds. A **head-to-head-direct** comparison on the same benchmark suite is planned for future work.

Baseline constraint handling. The black-box BO baseline uses a fixed penalty (10^6) for constraint violations. More sophisticated constrained BO methods—feasibility-weighted expected improvement [26], augmented Lagrangian approaches [42]—would likely narrow the gap on tightly constrained problems such as Heat-Exchanger and Distillation, where the penalty method **fails hardest** is **least effective**. However, the dimensionality reduction advantage of bilevel BO would persist regardless of constraint handling, as the surrogate dimension reduction from $n_{WB} + n_{BB}$ to n_{BB} is independent of the acquisition function. We intentionally use vanilla methods for both BO and NLP to isolate the effect of the bilevel structure rather than specific algorithmic choices.

Limitations and open problems. We identify six limitations, each pointing to an open problem. (1) The ~~method requires knowing the separable structure a priori; automatic detection of which variables are black-box versus white-box—perhaps via sensitivity analysis or computational graph inspection—would broaden applicability.~~ (2) The ~~inner NLP inner optimizer~~ must converge to the global optimum (Assumption 1(ii)). For non-convex inner problems, ~~multi-start solvers may find only local optima—this likely the inner solver may converge to local optima—as observed in Section 6, this explains the modest 1411× improvement on Rosen-Suzuki-SFR-1 (SLSQP variant) and the sub-100% feasibility on Batch-Reactor and Evaporator (Figure 7). Rigorous handling via deterministic global solvers Replacing the stochastic inner solver with a deterministic global solver (e.g., BARON) is future work.~~ (would provide optimality certificates but at higher computational cost. (2) The inner solver can perform *worse* than the black-box baseline on problems with very narrow feasible regions: on SFR-1, the BH inner solver’s random perturbations generate predominantly infeasible starting points, yielding a 0.1× ratio. Multi-start SLSQP avoids this failure mode on SFR-1 but lacks BH’s global exploration for multimodal problems. (3) Expensive inner ~~NLPs optimizations~~ can dominate wall time, ~~as~~. With the BH inner solver, Williams-Otto ~~demonstrates (111s requires 3,130s vs. 26s)273s for BB-BO (11.5× overhead); even the faster SLSQP variant incurs a 1.19× overhead on this problem.~~ Warm-starting the inner ~~NLP solver~~ from previous solutions or using reduced-order inner models could mitigate this. (4) Pure black-box constraints—those depending directly on f^{BB} outputs without passing through the white-box model—are not handled; extending the outer loop with constrained BO methods [26, 27] would address this. (5) All black-box functions in our suite are closed-form surrogates of what would be expensive computations; validation on a real DFT or molecular dynamics black box would strengthen the practical case. (6) Formal convergence guarantees for the outer BO loop—which follow from standard results since it is BO over a compact domain with a GP surrogate—are not analyzed here; characterizing how the bilevel structure affects dimension-dependent regret bounds is future work.

Broader implications. Expensive black-box optimization problems frequently contain exploitable structure—known physics, analytical submodels, or differentiable components—that monolithic surrogates ignore. The bilevel decomposition is one instance of a general principle: *solve what you can solve exactly, and ~~surrogate approximate~~ only what you must.* As surrogate-based methods scale to higher-dimensional search spaces, structural decomposition becomes not merely helpful but necessary to overcome the curse of dimensionality. The 13-problem benchmark suite provides a testbed for developing and comparing methods that exploit this structure.

8 Conclusion

When grey-box optimization problems exhibit variable separability—the expensive black-box depends only on one variable group while the remaining variables enter known, differentiable equations—solving the known subproblem exactly and ~~surrogating approximating~~ only the black-box component yields ~~1411×–277,000×–10⁸×~~ lower regret than monolithic surrogate-based optimization. The advantage ~~grows with dimension,~~ is robust to hyperparam-

eters [and inner-solver choice](#), and requires no algorithmic novelty beyond the decomposition itself. A suite of 13 benchmark problems with verified global optima provides a standardized testbed for future methods that exploit this structure.

The opening observation of this paper—that monolithic surrogates waste samples learning what is already known—is nearly self-evident once the separable structure is recognized. Yet no prior method systematically exploits this structure for dimensionality reduction with exact constraint satisfaction (in principle; subject to inner-solver global optimality in practice). The empirical evidence across [78,410–450](#) independent optimization runs confirms that the payoff is large and consistent. As expensive black-box optimization scales to higher dimensions, identifying and exploiting known substructure will be essential; the bilevel decomposition studied here is one principled way to do so.

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Declarations

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- **Conflict of interest.** The authors declare that they have no conflict of interest.
- **Data availability.** All data generated during this study are reproducible from the code described below. No external datasets were used.
- **Code availability.** The Python code for the benchmark suite, solvers, and experiment runner is available at <https://github.com/joshuaeh/HybridBayesianOptimization>.
- **Author contributions.** Joshua E. Hammond: conceptualization, methodology, software, investigation, writing—original draft. Tyler A. Soderstrom: methodology, investigation, writing—review & editing. Brian A. Korgel: supervision, writing—review & editing. Michael Baldea: conceptualization, supervision, funding acquisition, writing—review & editing.
- **Use of generative AI.** Generative AI tools (Claude Opus 4.6, Anthropic) were used to assist with code development, data analysis, and manuscript drafting. All AI-generated content was reviewed, verified, and edited by the authors, who take full responsibility for the accuracy and integrity of the work. No AI tool was used to generate scientific claims, interpret results, or design experiments without human oversight.

Appendix A Test Problem Definitions

Full mathematical formulations for all 13 benchmark problems. Each problem specifies the white-box and black-box models, constraints, variable bounds, and verified global optimum.

A.1 Small Feasible Region Problem 1

Gardner et al. [26] and Ariaifar et al. [39] propose a constrained optimization test problem with a small feasible region that can be formulated as follows:

$$\begin{aligned} \min_x \quad & \sin x_1 + y_1, \\ \text{s.t.} \quad & g_1(x) := \sin(x_1) \sin(y_1) + 0.95 \leq 0, \\ & y_1 = f^{\text{BB}}(x) := x_2, \\ & 0 \leq x_i \leq 6, \quad \forall i \in \{1, 2\} \end{aligned} \tag{6}$$

The global minimum is ~~$J^* = -0.253236$~~ $J^* = 0.2532$ and is located at ~~$x_1 = \frac{3\pi}{2}$~~ , ~~$x^{\text{BB}} = x_2 = 0.253236$~~ $x_1^{\text{WB}^*} = \frac{3\pi}{2}$, $x^{\text{BB}^*} = x_2 = 1.2532$.

A.2 Modified Small Feasible Region Problem 2

Gardner et al. [26] and Ariaifar et al. [39] propose a constrained optimization test problem with a small feasible region. We add an additional black-box function $y_1 = f^{\text{BB}}(x_2)$ that forms a grey-box problem as follows:

$$\begin{aligned} \min_x \quad & \cos(2x_1) \cos(y_1) + \sin(x_1), \\ \text{s.t.} \quad & g_1(x) := \cos(x_1) \cos(y_1) - \sin(x_1) \sin(y_1) - 0.5 \leq 0, \\ & y_1 = f^{\text{BB}}(x) := \frac{(x_2 - 5.5)}{2} \exp\left(\frac{x_2}{2}\right), \\ & 0 \leq x_i \leq 6, \quad \forall i \in \{1, 2\} \end{aligned} \tag{7}$$

The global minimum $J^* = -2.0$ is located at $x^{\text{BB}^*} = 5.5, x^{\text{WB}^*} = 3\pi/2$.

A.3 Rastrigin

We use a modified formulation of the Rastrigin function [43] that can be formulated as a multi-scale grey-box problem as follows:

$$\begin{aligned} \min_{x,y} \quad & 30 + x_1^2 - 10 \cos(2\pi x_1) + x_2^2 - 10 \cos(2\pi x_2) + y_1, \\ \text{s.t.} \quad & y_1 = f^{\text{BB}}(x) := x_3^2 - 10 \cos(2\pi x_3), \\ & -5.12 \leq x_i \leq 5.12, \quad \forall i \in \{1, 2, 3\}, \\ & x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [x_1, x_2], \quad x^{\text{BB}} = [x_3]. \end{aligned} \tag{8}$$

The global minimum is equal to 0 with $x^{\text{WB}^*} = [0, 0], x^{\text{BB}^*} = [0]^\top$.

A.4 Toy-Hydrology

We use a modified formulation of the Toy Hydrology problem [42] that can be formulated as a multi-scale grey-box problem as follows:

$$\begin{aligned}
& \min_{x,y} x_1 + x_2, \\
& \text{s.t. } g_1(x, y) := 1.5 - x_1 - 2x_2 - 0.5 \sin(-4\pi x_2 + y_1) \leq 0, \\
& \quad g_2(x, y) := x_1^2 + x_2^2 - 1.5 \leq 0, \\
& \quad y_1 = f^{\text{BB}}(x_1) := 2\pi x_1^2, \\
& \quad 0 \leq x_i \leq 1, \quad \forall i \in \{1, 2\}, \\
& \quad x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{BB}} = [x_1], \quad x^{\text{WB}} = [x_2].
\end{aligned} \tag{9}$$

The global minimum is 0.5998 with $x^{\text{WB}\star} = 0.1951, x^{\text{BB}\star} = 0.4047, x^{\text{WB}\star} = 0.4047, x^{\text{BB}\star} = 0.1951$.

A.5 Rosen-Suzuki

We use a modified formulation of the Rosen-Suzuki function [44] that can be formulated as a multi-scale grey-box problem as follows:

$$\begin{aligned}
& \min_{x,y} x_1^2 + x_2^2 + x_4^2 - 5x_1 - 5x_2 + y_1, \\
& \text{s.t. } g_1(x, y) := -(8 - x_1^2 - x_2^2 - x_3^2 - x_4^2 - x_1 + x_2 - x_3 + x_4) \leq 0, \\
& \quad g_2(x, y) := -(10 - x_1^2 - 2x_2^2 - y_2 + x_1 + x_4) \leq 0, \\
& \quad g_3(x, y) := -(5 - 2x_1^2 - x_2^2 - x_3^2 - 2x_1 + x_2 + x_4) \leq 0, \\
& \quad y_1 = f_1^{\text{BB}}(z) := 2x_3^2 - 21x_3 + 7x_4, \\
& \quad y_2 = f_2^{\text{BB}}(z) := x_3^2 + 2x_4^2, \\
& \quad -2 \leq x_i \leq 2, \quad \forall i \in \{1, \dots, 4\} \\
& \quad x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [x_1, x_2], \quad x^{\text{BB}} = [x_3, x_4].
\end{aligned} \tag{10}$$

The global minimum is -44 with $x^{\text{WB}\star} = [0, 1]^\top, x^{\text{BB}\star} = [2, -1]^\top$.

A.6 Catalytic Reactor Design

We use a catalytic reactor design problem that combines a white-box CSTR process model with black-box catalyst properties [akin to those](#) derived from DFT calculations. The objective is to *maximize* the yield of intermediate product B in consecutive reactions $A \rightarrow B \rightarrow C$. The problem can be formulated

as a multi-scale grey-box problem as follows:

$$\begin{aligned}
& \max_{T, P, \tau, E_b, \Delta E_a} Y_B \tag{11} \\
& \text{s.t. } \mathbf{f}^{\text{WB}} = \begin{bmatrix} Y_B - \frac{k_1 \tau}{(1+k_1 \tau)(1+k_2 \tau)} \\ k_1(T, P, E_b, \Delta E_a) - A_1 v(E_b) \exp\left(-\frac{E_{a,1}}{RT}\right) \left(\frac{P}{P_{\text{ref}}}\right)^{\beta_1} \frac{1}{(1+K_{\text{ads}} P)^2} \\ k_2(T, P, E_b, \Delta E_a) - A_2 \exp\left(-\frac{E_{a,2}}{RT}\right) \left(\frac{P}{P_{\text{ref}}}\right)^{\beta_2} \frac{1}{(1+K_{\text{ads}} P)^2} \\ K_{\text{ads}}(E_b) - K_0 \exp(\kappa(E_b - E_b^*)) \end{bmatrix}, \\
& \mathbf{f}^{\text{BB}} = \begin{bmatrix} v(E_b) - \left[\exp\left(-\frac{1}{2} \left(\frac{E_b - E_b^*}{\sigma_E}\right)^2\right) \right] \\ E_{a,1} - \left[E_{a,1}^{(0)} - \alpha_1 \Delta E_a \right] \\ E_{a,2} - \left[E_{a,2}^{(0)} + \alpha_2 \Delta E_a \right] \end{bmatrix}, \\
& g_1(x, y) := Y_C - 0.25 \leq 0, \\
& g_2(x, y) := \frac{P}{P_{\text{max}}} - \frac{T - T_{\text{min}}}{T_{\text{max}} - T_{\text{min}}} - 0.3 \leq 0,
\end{aligned}$$

and the conversion and byproduct yield are:

$$X_A = \frac{k_1 \tau}{1 + k_1 \tau}, \quad Y_C = X_A - Y_B.$$

The black-box function f^{BB} maps catalyst properties ΔE_a , E_b to kinetic parameters via a volcano-type activity model:

The global maximum yield [given the parameters in Table 4](#) is 92.32% with $x^{\text{WB}\star} = [647.25, 20.00, 5.00]^\top$, $x^{\text{BB}\star} = [-1.047, 0.20]^\top$, $x^{\text{WB}\star} = [647.3, 20.0, 5.0]^\top$, $x^{\text{BB}\star} = [-1.047, 0.200]^\top$.

A.7 Heat Exchanger with Novel Working Fluid

We consider the design of a heat exchanger system where the working fluid's thermodynamic properties (heat capacity c_p , viscosity μ , and thermal conductivity k) depend on two molecular descriptors (m_1 , m_2) in a black-box function. White-box equations govern the heat exchanger design and operation (heat transfer area A , mass flow rate $\dot{m}$, and log-mean temperature difference ΔT_{lm}) with the goal of minimizing the total cost (capital + operating) while satisfying minimum heat duty [45]. Capital cost scales with area according to the six-tenths rule [46] while operating cost is a function of pumping power which depends on fluid viscosity and mass flow rate according to the relation provided by [47]. We use a simplified form for the overall heat transfer coefficient U based the resistance model from Ravagnani and Caballero [48].

Table 4: Parameters for the catalytic reactor design problem.

Symbol	Description	Range / Value	Units
Decision / design variables			
T	Temperature	[450, 700]	K
P	Pressure	[1, 20]	bar
τ	Residence time	[0.1, 5.0]	s
E_b	Binding energy	[-2.0, 0.0]	eV
ΔE_a	Activation-energy shift	[-0.2, 0.2]	-
A_1	Pre-exponential (step 1)	1.0×10^7	s^{-1}
A_2	Pre-exponential (step 2)	3.0×10^6	s^{-1}
$E_{a1}(\Delta E_a)$	Activation energy	$80,000 - 20,000\Delta E_a$	J/mol
$E_{a2}(\Delta E_a)$	Activation energy	$95,000 + 5,000\Delta E_a$	J/mol
P_{ref}	Reference pressure	8.0	bar
E_b^*	Reference binding energy	-1.0	eV
σ_E	Width parameter	0.40	eV
K_0	Adsorption rate pre-exponential factor	0.06	1/bar
κ	Adsorption rate exponential term	3.0	1/eV
R	Gas constant	8.314	J/(mol K)
k_B	Boltzmann constant	8.617×10^{-5}	eV/K
α_1		20,000	-
α_2		5,000	-
β_1	Exponent for pressure dependence	1.0	-
β_2	Exponent for pressure dependence	0.35	-

The problem can be formulated as:

$$\begin{aligned}
& \min_{A, \dot{m}, \Delta T_{lm}, m_1, m_2} C_{\text{capital}} + C_{\text{operating}} & (12) \\
& \text{s.t. } C_{\text{capital}} = 1000 \cdot x_1^{\text{WB}0.6}, \\
& C_{\text{operating}} = 500 \cdot \frac{x_2^{\text{WB}3} \cdot y_2}{x_1^{\text{WB}}}, \\
& U = \frac{y_3}{0.01 + 0.001/y_1}, \\
& g_1(x, y) := 100 - U \cdot x_1^{\text{WB}} \cdot x_3^{\text{WB}} \leq 0, \\
& g_2(x, y) := 4000 - \frac{4x_2^{\text{WB}}}{\pi \cdot 0.05 \cdot y_2} \leq 0, \\
& y_1 = f_1^{\text{BB}}(x_1^{\text{BB}}, x_2^{\text{BB}}) := 2.0 + 0.5 \sin\left(\frac{\pi x_1^{\text{BB}}}{50}\right) + 0.3x_2^{\text{BB}}, \\
& y_2 = f_2^{\text{BB}}(x_1^{\text{BB}}, x_2^{\text{BB}}) := 0.001 \cdot \exp\left(\frac{x_1^{\text{BB}} - 100}{50}\right) \cdot (1 + 0.5x_2^{\text{BB}}), \\
& y_3 = f_3^{\text{BB}}(x_1^{\text{BB}}, x_2^{\text{BB}}) := 0.1 + 0.05 \left(1 - \left(\frac{x_1^{\text{BB}} - 125}{75}\right)^2\right) + 0.02x_2^{\text{BB}}, \\
& A \in [1, 50], \quad \dot{m} \in [0.1, 5], \quad \Delta T_{lm} \in [5, 50], \\
& m_1 \in [50, 200], \quad m_2 \in [0, 1], \\
& x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [A, \dot{m}, \Delta T_{lm}], \quad x^{\text{BB}} = [m_1, m_2],
\end{aligned}$$

Table 5: Variables, physical meaning, bounds, and units (Heat exchanger)

Variable	Physical meaning	Lower	Upper	Units
x_1^{WB}	Heat transfer area (A)	1	50	m ²
x_2^{WB}	Mass flow rate ($\dot{m}$)	0.1	5	kg/s
x_3^{WB}	Log-mean temperature difference (ΔT_{lm})	5	50	K
x_1^{BB}	Molecular descriptor (m_1)	50	200	(unitless)
x_2^{BB}	Molecular descriptor (m_2)	0	1	(unitless)
y_1	Heat capacity (c_p)	N/A	N/A	kJ/kg·K
y_2	Viscosity (μ)	N/A	N/A	Pa·s
y_3	Thermal conductivity (k)	N/A	N/A	W/m·K

where y_1 is the heat capacity [kJ/kg·K], y_2 is the viscosity [Pa·s], and y_3 is the thermal conductivity [W/m·K]. The constraint g_1 ensures the heat duty $Q = U \cdot A \cdot \Delta T_{lm} \geq 100$ kW is satisfied, and g_2 enforces turbulent flow with Reynolds number $\text{Re} \geq 4000$. The global minimum cost [given the parameters in Table 5](#) is 1000.0 with $x^{\text{WB}\star} = [1.0, 0.1, 39.59]^\top$, $x^{\text{BB}\star} = [1.0, 0.1, 19.3]^\top$, $x^{\text{BB}\star} = [50.0, 0.0]^\top$.

A.8 Pressure Swing Adsorption with Novel Adsorbent

We consider the design of a pressure swing adsorption (PSA) cycle for gas separation, a problem that naturally exhibits grey-box structure [3, 49, 50]. The adsorbent properties—maximum loading q_{max} , adsorption equilibrium constant

for the target component K_A , and equilibrium constant for the competing species K_B —are computed from molecular simulations [51, 52] that serve as black-box functions of adsorbent structural parameters (pore size σ and interaction energy ϵ) [53].

The white-box model captures the PSA cycle performance through Langmuir isotherm-based working capacity calculations [54, 55]. The working capacity Δq represents the difference in loading between high-pressure adsorption and low-pressure desorption steps [56, 50]. Selectivity $\alpha = K_A/K_B$ determines separation performance [57], while productivity is defined as working capacity divided by adsorption time [58]. The objective minimizes the trade-off between maximizing productivity and minimizing compression costs, where isothermal compression work scales with $\ln(P_H/P_L)$ [59].

The problem can be formulated as:

$$\begin{aligned}
\min_{P_H, P_L, t_{\text{ads}}, \sigma, \epsilon} \quad & -\text{Prod} + 0.1 \cdot \ln\left(\frac{P_H}{P_L}\right) \\
\text{s.t.} \quad & \alpha = \frac{y_2}{y_3}, \\
& \Delta q = y_1 \cdot \left(\frac{y_2 \cdot P_H}{1 + y_2 \cdot P_H} - \frac{y_2 \cdot P_L}{1 + y_2 \cdot P_L} \right), \\
& \text{Prod} = \frac{\Delta q}{t_{\text{ads}}}, \\
& g_1(x, y) := 3 - \alpha \leq 0, \\
& g_2(x, y) := 0.95 - \frac{\alpha \cdot P_H}{1 + \alpha \cdot P_H} \leq 0, \\
& y_1 = f_1^{\text{BB}}(\sigma, \epsilon) := 5.0 \cdot \exp\left(-\frac{(\sigma - 6)^2}{4}\right) \cdot (1 + 0.1\epsilon), \\
& y_2 = f_2^{\text{BB}}(\sigma, \epsilon) := 0.5 \cdot \exp\left(\frac{\epsilon - 10}{5}\right) \cdot (1 - 0.05(\sigma - 5)^2), \\
& y_3 = f_3^{\text{BB}}(\sigma, \epsilon) := 0.1 \cdot \exp\left(\frac{\epsilon - 15}{8}\right) \cdot (1 + 0.03(\sigma - 7)^2), \\
& P_H \in [2, 10], \quad P_L \in [0.1, 1], \quad t_{\text{ads}} \in [10, 120], \\
& \sigma \in [3, 10], \quad \epsilon \in [5, 25], \\
& x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [P_H, P_L, t_{\text{ads}}], \quad x^{\text{BB}} = [\sigma, \epsilon],
\end{aligned} \tag{13}$$

The black-box functions f^{BB} model the adsorbent properties as functions of the Lennard-Jones parameters: pore size σ [Å] and interaction energy ϵ [kJ/mol] [53, 52]. These relationships capture how molecular-scale adsorbent design affects macroscopic separation performance [51].

The constraint g_1 ensures minimum selectivity $\alpha \geq 3$ for effective separation [57, 56], and g_2 enforces minimum product purity based on the Langmuir isotherm equilibrium [54, 50]. The global minimum [given the parameters in Table 6](#) is -0.6433 with $x^{\text{WB}\star} = [3.97, 0.1, 10.0]^\top$, $x^{\text{BB}\star} = [6.04, 19.55]^\top$.

A.9 Batch Reactor with Catalyst Optimization

We consider the optimization of a batch reactor for consecutive reactions $A \rightarrow B \rightarrow C$, where the objective is to maximize the yield of the intermediate product B [60]. This classic selectivity problem [61] gains a multi-scale

Table 6: Variables, physical meaning, bounds, and units (Pressure Swing Adsorption)

Symbol	Description	Range / Value	Units
White-box decision variables			
P_H	High (adsorption) pressure	[2, 10]	bar
P_L	Low (desorption) pressure	[0.1, 1]	bar
t_{ads}	Adsorption step time	[10, 120]	s
Black-box decision variables			
σ	Adsorbent pore size	[3, 10]	Å
ϵ	Lennard-Jones interaction energy	[5, 25]	kJ/mol
Black-box outputs			
$y_1 = q_{\text{max}}$	Maximum loading capacity	—	mol/kg
$y_2 = K_A$	Adsorption equilibrium constant (component A)	—	1/bar
$y_3 = K_B$	Adsorption equilibrium constant (component B)	—	1/bar

dimension when catalyst properties must be co-optimized with operating conditions.

The white-box model consists of standard Arrhenius kinetics [62] governing the batch reactor mass balances, which admit analytical solutions for first-order consecutive reactions. Temperature T , batch time t_f , and initial concentration $C_{A,0}$ are the white-box decision variables.

The black-box model implements a Sabatier volcano relationship [63, 64, 65] that captures how catalyst activity depends on binding energy E_b . The volcano principle states that optimal catalysts bind reactants neither too strongly (limiting desorption) nor too weakly (limiting adsorption), with peak activity at an intermediate binding energy. The catalyst particle size d provides an additional degree of freedom that affects the pre-exponential factor through available surface area.

The problem can be formulated as:

$$\begin{aligned}
& \min_{T, t_f, C_{A,0}, E_b, d} && -C_B + 0.001 \cdot T \cdot t_f && (14) \\
& \text{s.t.} && k'_1 = y_1 \cdot \exp\left(\frac{-2000}{T}\right), \\
& && k'_2 = y_2 \cdot \exp\left(\frac{-2400}{T}\right), \\
& && C_A = C_{A,0} \cdot \exp(-k'_1 \cdot t_f), \\
& && C_B = C_{A,0} \cdot \frac{k'_1}{k'_2 - k'_1} \cdot (\exp(-k'_1 \cdot t_f) - \exp(-k'_2 \cdot t_f)), \\
& && g_1(x, y) := 0.8 - \left(1 - \frac{C_A}{C_{A,0}}\right) \leq 0, \\
& && g_2(x, y) := 0.7 - \frac{C_B}{C_{A,0} - C_A} \leq 0, \\
& && y_1 = f_1^{\text{BB}}(E_b, d) := 10 \cdot d \cdot \exp\left(-\frac{(E_b + 1)^2}{0.25}\right), \\
& && y_2 = f_2^{\text{BB}}(E_b, d) := 0.5 \cdot d \cdot \exp\left(-\frac{(E_b + 0.5)^2}{0.5}\right), \\
& && y_3 = f_3^{\text{BB}}(E_b, d) := \exp(-2 \cdot E_b - 1), \quad (\text{not used in white-box equations}) \\
& && T \in [300, 500], \quad t_f \in [0.5, 10], \quad C_{A,0} \in [0.5, 5], \\
& && E_b \in [-2, 0], \quad d \in [0.5, 2], \\
& && x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [T, t_f, C_{A,0}], \quad x^{\text{BB}} = [E_b, d],
\end{aligned}$$

The Gaussian functions in f_1^{BB} and f_2^{BB} encode the volcano-shaped activity curves: k_1 peaks at $E_b = -1$ eV while k_2 peaks at $E_b = -0.5$ eV, creating a selectivity landscape where catalyst design can favor the desired reaction over the consecutive side reaction. The output y_3 does not appear in any white-box equation or constraint; it is included intentionally to test robustness to irrelevant black-box outputs, since in practice molecular simulations often return quantities not all of which enter the process model.

The constraint g_1 ensures minimum conversion $X_A \geq 0.8$, and g_2 enforces selectivity $S_B = C_B/(C_{A,0} - C_A) \geq 0.7$, requiring that at least 70% of converted reactant forms the desired intermediate. The global minimum ~~is -1.7485 given the parameters in Table 7~~ is -1.7488 with $x^{\text{WB}\star} = [500.0, 4.394, 5.0]^\top$, ~~$x^{\text{BB}\star} = [-1.0, 2.0]^\top$~~ $x^{\text{BB}\star} = [-1.01, 2.0]^\top$.

A.10 Extractive Distillation with Novel Solvent

We consider the integrated design of an extractive distillation column and entrainer selection, a canonical example of simultaneous solvent and process optimization [66, 67]. The white-box model employs classical shortcut methods: the Fenske equation [68] determines minimum stages $N_{\min}$ from the relative volatility and separation specifications, the Underwood equations [69, 70] yield minimum reflux $R_{\min}$, and operating constraints follow from the Gilliland correlation [71, 72].

The black-box model represents COSMO-RS or other molecular thermodynamic predictions that map solvent molecular descriptors—here parameterized by Hansen solubility parameters for hydrogen bonding δ_H and polar

Table 7: Variables, physical meaning, bounds, and units (Batch Reactor)

Symbol	Description	Range / Value	Units
White-box decision variables			
T	Reactor temperature	[300, 500]	K
t_f	Batch (final) time	[0.5, 10]	h
$C_{A,0}$	Initial concentration of A	[0.5, 5]	mol/L
Black-box decision variables			
E_b	Catalyst binding energy	[-2, 0]	eV
d	Catalyst particle size factor	[0.5, 2]	—
Black-box outputs			
$y_1 = k_1^0$	Pre-exponential factor for $A \rightarrow B$	—	1/s
$y_2 = k_2^0$	Pre-exponential factor for $B \rightarrow C$	—	1/s
$y_3 = K_{eq}$	Adsorption equilibrium constant	—	—

interactions δ_P —to the modified relative volatility α_{12}^{mod} in the presence of the entrainer [73]. The entrainer also affects operating costs through its density ρ (determining column sizing) and viscosity μ (affecting tray hydraulics and heat transfer).

The problem can be formulated as:

$$\begin{aligned}
 \min_{R, N, S/F, \delta_H, \delta_P} \quad & 1000 \cdot N + 500 \cdot R + 200 \cdot (S/F) \cdot y_3 \quad (15) \\
 \text{s.t.} \quad & R_{\min} = \frac{1}{y_1 - 1} \cdot \left(\frac{x_D}{x_F} - y_1 \cdot \frac{1 - x_D}{1 - x_F} \right), \\
 & N_{\min} = \frac{\ln \left(\frac{x_D(1-x_B)}{x_B(1-x_D)} \right)}{\ln(y_1)}, \\
 & g_1(x, y) := 1.2 \cdot R_{\min} - R \leq 0, \\
 & g_2(x, y) := 1.5 \cdot N_{\min} - N \leq 0, \\
 & g_3(x, y) := (S/F) \cdot y_2 - 5000 \leq 0, \\
 & y_1 = f_1^{\text{BB}}(\delta_H, \delta_P) := 2.5 + 1.0 \cdot \sin \left(\frac{\pi(\delta_H - 10)}{20} \right) \cdot \cos \left(\frac{\pi(\delta_P - 10)}{15} \right), \\
 & y_2 = f_2^{\text{BB}}(\delta_H, \delta_P) := 800 + 50 \cdot \delta_H - 20 \cdot \delta_P, \\
 & y_3 = f_3^{\text{BB}}(\delta_H, \delta_P) := 0.5 + 0.1 \cdot \delta_H + 0.05 \cdot \delta_P^2/100, \\
 & R \in [1, 10], \quad N \in [5, 50], \quad S/F \in [0.5, 5], \\
 & \delta_H \in [5, 25], \quad \delta_P \in [5, 20], \\
 & x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [R, N, S/F], \quad x^{\text{BB}} = [\delta_H, \delta_P],
 \end{aligned}$$

where $x_D = 0.99$ (distillate purity), $x_F = 0.5$ (feed composition), and $x_B = 0.01$ (bottoms impurity).

The constraint g_1 ensures the reflux ratio exceeds the minimum by at least 20% for operational flexibility, g_2 requires sufficient stages above the Fenske minimum, and g_3 limits solvent inventory based on density to control capital costs. The global minimum cost ~~is 11759.0 for the parameters in Table 8~~ is 11.758 with $x^{\text{WB}\star} = [1.0, 11.0, 0.5]^\top$, ~~$x^{\text{BB}\star} = [20.0, 10.0]^\top$~~ $x^{\text{BB}\star} = [19.8, 9.99]^\top$.

Table 8: Variables, physical meaning, bounds, and units (Extractive Distillation)

Symbol	Description	Range / Value	Units
White-box decision variables			
R	Reflux ratio	[1, 10]	—
N	Number of theoretical stages	[5, 50]	—
S/F	Solvent-to-feed ratio	[0.5, 5]	—
Black-box decision variables			
δ_H	Hansen hydrogen-bonding parameter	[5, 25]	MPa ^{0.5}
δ_P	Hansen polar parameter	[5, 20]	MPa ^{0.5}
Black-box outputs			
$y_1 = \alpha_{12}^{\text{mod}}$	Modified relative volatility	—	—
$y_2 = \rho$	Solvent density	—	kg/m ³
$y_3 = \mu$	Solvent viscosity	—	cP
Fixed parameters			
x_D	Distillate purity specification	0.99	mol/mol
x_F	Feed composition	0.50	mol/mol
x_B	Bottoms impurity specification	0.01	mol/mol

A.11 Multi-Effect Evaporator with Phase-Change Material

We consider a ~~stylized-prototype~~ test problem for the design of a multi-effect evaporator (MEE) system integrated with a novel phase-change material (PCM) as the heat transfer medium. Multi-effect evaporator synthesis follows established energy balance formulations [74], where steam economy improves with additional effects but with diminishing returns due to ~~easeading-tighter~~ temperature driving forces. The efficiency degradation across effects is captured by the factor 0.9^{n-1} , consistent with typical industrial values of 0.7–0.9 pounds of evaporation per pound of steam per effect [75].

The white-box model includes energy balances relating the heat transfer fluid (HTF) flow rate and properties to evaporation capacity [76]. The latent heat of vaporization for water (2260 kJ/kg) determines the evaporation rate from available thermal energy. The number of effects n , HTF mass flow rate $\dot{m}_{\text{HTF}}$, and temperature drop per effect ΔT_{effect} are the white-box decision variables.

The black-box model represents molecular simulation or group-contribution predictions of PCM thermophysical properties [77, 78] as functions of melting temperature T_m and a molecular structure parameter λ that controls the length of alkyl chains (affecting both latent heat and heat capacity). The key outputs are latent heat of fusion ΔH_{fus} (typical range 150–300 kJ/kg for organic PCMs), liquid heat capacity c_p^{liq} (typically 2–3 kJ/(kg·K)), and the actual melting temperature T_{melt} . The functional forms in f^{BB} are ~~stylized representations that intended to~~ capture qualitative structure-property relationships rather than precise physical models.

The problem can be formulated as:

$$\begin{aligned}
& \min_{n, \dot{m}_{\text{HTF}}, \Delta T_{\text{effect}}, T_m, \lambda} && -\dot{m}_{\text{evap}} + 0.1 \cdot n^2 + 0.01 \cdot \dot{m}_{\text{HTF}}^2 && (16) \\
& \text{s.t.} && Q_{\text{HTF}} = \dot{m}_{\text{HTF}} \cdot (y_1 + y_2 \cdot 10), \\
& && \dot{m}_{\text{evap}} = \frac{Q_{\text{HTF}} \cdot n \cdot 0.9^{n-1}}{2260}, \\
& && g_1(x, y) := 100 + n \cdot \Delta T_{\text{effect}} - y_3 \leq 0, \\
& && g_2(x, y) := 5 - \dot{m}_{\text{evap}} \leq 0, \\
& && y_1 = f_1^{\text{BB}}(T_m, \lambda) := 150 \cdot \lambda \cdot \left(1 + 0.2 \sin\left(\frac{\pi T_m}{100}\right)\right), \\
& && y_2 = f_2^{\text{BB}}(T_m, \lambda) := 2.0 + 0.5 \cdot \lambda - 0.002 \cdot T_m, \\
& && y_3 = f_3^{\text{BB}}(T_m, \lambda) := T_m + 10 \cdot \sin\left(\frac{\pi \lambda}{2}\right), \\
& && n \in [2, 6] \text{ (treated as continuous; integer rounding would apply in practice),} \\
& && \dot{m}_{\text{HTF}} \in [1, 20], \quad \Delta T_{\text{effect}} \in [5, 20], \\
& && T_m \in [50, 150], \quad \lambda \in [0.5, 2], \\
& && x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [n, \dot{m}_{\text{HTF}}, \Delta T_{\text{effect}}], \quad x^{\text{BB}} = [T_m, \lambda],
\end{aligned}$$

Table 9: Variables, physical meaning, bounds, and units (Multi-Effect Evaporator)

Symbol	Description	Range / Value	Units
White-box decision variables			
n	Number of evaporator effects	[2, 6]	—
$\dot{m}_{\text{HTF}}$	Heat transfer fluid mass flow rate	[1, 20]	kg/s
ΔT_{effect}	Temperature drop per effect	[5, 20]	°C
Black-box decision variables			
T_m	Target melting temperature	[50, 150]	°C
λ	Molecular structure parameter	[0.5, 2]	—
Black-box outputs			
$y_1 = \Delta H_{\text{fus}}$	Latent heat of fusion	—	kJ/kg
$y_2 = c_p^{\text{liq}}$	Liquid phase heat capacity	—	kJ/(kg·K)
$y_3 = T_{\text{melt}}$	Actual melting temperature	—	°C
Fixed parameters			
$\Delta H_{\text{vap,water}}$	Latent heat of water vaporization	2260	kJ/kg
η_{effect}	Per-effect efficiency factor	0.9	—

The constraint g_1 ensures temperature feasibility: the PCM melting temperature must exceed the cumulative temperature drop across all effects plus a 100°C baseline. The constraint g_2 enforces a minimum evaporation rate of 5 kg/s to meet production requirements. The global minimum [for parameters in Table 9](#) is -1.9587 with $x^{\text{WB}\star} = [4.09, 19.07, 5.0]^\top$, $x^{\text{BB}\star} = [120.47, 2.0]^\top$.

A.12 Membrane Separation with Designed Polymer

We consider the integrated design of a membrane gas separation process and the polymer membrane material [79]. This problem exemplifies the permeability-selectivity trade-off characterized by the Robeson upper bound [80, 81], which establishes that polymeric membranes exhibit a fundamental inverse relationship between gas permeability and selectivity explained by free volume theory [82].

The white-box model follows the solution-diffusion transport mechanism [83], where the permeate flux J_A is proportional to permeability, pressure driving force, and inversely proportional to membrane thickness δ . The membrane area A_m , thickness δ , and transmembrane pressure difference Δp are the white-box decision variables, with the objective of maximizing product recovery while minimizing capital (area) and operating (compression) costs.

The black-box model represents structure-property relationships from molecular simulations or machine learning predictions [84] that map polymer descriptors—fractional free volume (FFV) and chain spacing d_{spacing} —to transport properties. The Robeson trade-off is encoded in the black-box functions: increasing FFV enhances permeability but reduces selectivity, while chain spacing affects both the upper bound position and mechanical integrity. ~~Process superstructure considerations motivate the~~ The mechanical strength constraint reflects the requirement that the membrane must withstand the transmembrane pressure difference during module operation [85].

The problem can be formulated as:

$$\begin{aligned}
 \min_{A_m, \delta, \Delta p, \text{FFV}, d_{\text{spacing}}} \quad & -F_A \cdot y_2 + 0.1 \cdot A_m + 10 \cdot \Delta p \tag{17} \\
 \text{s.t.} \quad & J_A = \frac{y_1 \cdot \Delta p}{\delta} \cdot 10^{-10}, \\
 & F_A = J_A \cdot A_m, \\
 & g_1(x, y) := 10^{-6} - J_A \leq 0, \\
 & g_2(x, y) := \frac{\Delta p \cdot A_m}{y_3} - 100 \leq 0, \\
 & y_1 = f_1^{\text{BB}}(\text{FFV}, d_{\text{spacing}}) := 1000 \cdot \exp\left(\frac{\text{FFV} - 0.15}{0.05}\right) \cdot (1 + 0.1 \cdot d_{\text{spacing}}), \\
 & y_2 = f_2^{\text{BB}}(\text{FFV}, d_{\text{spacing}}) := 50 \cdot \exp\left(-\frac{\text{FFV} - 0.15}{0.1}\right) \cdot \exp\left(-\frac{(d_{\text{spacing}} - 5)^2}{4}\right), \\
 & y_3 = f_3^{\text{BB}}(\text{FFV}, d_{\text{spacing}}) := 100 \cdot (0.4 - \text{FFV}) \cdot (10 - d_{\text{spacing}}), \\
 & A_m \in [10, 1000], \quad \delta \in [0.1, 10], \quad \Delta p \in [1, 20], \\
 & \text{FFV} \in [0.1, 0.3], \quad d_{\text{spacing}} \in [3, 8], \\
 & x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [A_m, \delta, \Delta p], \quad x^{\text{BB}} = [\text{FFV}, d_{\text{spacing}}],
 \end{aligned}$$

Refer to the parameter definitions in Table 10.

The white-box flux equation $J_A = y_1 \cdot \Delta p / \delta$ follows directly from the solution-diffusion model [83], where permeate flux is proportional to permeability and pressure driving force, and inversely proportional to membrane thickness. The factor 10^{-10} converts from Barrer (the conventional permeability unit) to SI flux units.

The black-box functions are constructed to capture established structure-property relationships from polymer physics. The permeability function y_1

Table 10: Variables, physical meaning, bounds, and units (Membrane Separation)

Symbol	Description	Range / Value	Units
White-box decision variables			
A_m	Membrane area	[10, 1000]	m ²
δ	Membrane thickness	[0.1, 10]	μm
Δp	Transmembrane pressure difference	[1, 20]	bar
Black-box decision variables			
FFV	Fractional free volume	[0.1, 0.3]	—
d_{spacing}	Polymer chain spacing	[3, 8]	$\text{\AA}$
Black-box outputs			
$y_1 = P_A$	Permeability of component A	—	Barrer
$y_2 = \alpha_{A/B}$	Selectivity (A over B)	—	—
$y_3 = \sigma$	Mechanical strength proxy	—	MPa

follows the exponential free volume dependence derived by Cohen and Turnbull [86] and extended by Fujita [87], who showed that diffusivity (and hence permeability) scales as $P \propto \exp(B \cdot \text{FFV})$ where B is a polymer-penetrant parameter. This exponential form arises because molecular transport occurs through transient gaps in the polymer matrix, with the probability of sufficiently large gaps increasing exponentially with free volume fraction [88]. The chain spacing term provides a secondary contribution reflecting the effect of interchain distance on diffusion pathways [89].

The selectivity function y_2 encodes the Robeson upper bound trade-off [80, 81]: as FFV increases, permeability rises but selectivity decreases because larger free volume elements become less size-discriminating. Freeman [82] provided a theoretical basis showing that the slope of the permeability-selectivity trade-off depends only on penetrant size ratios. The Gaussian dependence on d_{spacing} reflects an optimal chain spacing for size-sieving selectivity [85].

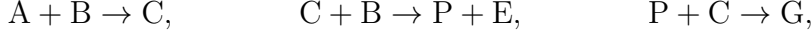
The mechanical strength proxy y_3 captures the physical principle that increased free volume and chain spacing reduce polymer density and chain entanglement, thereby weakening mechanical integrity [90]. While the linear functional form is a simplification, it correctly represents the inverse relationship between transport-enhancing microstructure and mechanical robustness that constrains practical membrane design [91].

The constraint g_1 ensures minimum flux $J_A \geq 10^{-6} \text{ mol}/(\text{m}^2 \cdot \text{s})$ for practical separation rates, and g_2 enforces mechanical stability by limiting the pressure-area product relative to membrane strength.

The global optimum ~~, verified via multi-start NLP with 1000 starting points, for parameters in Table 10~~ occurs at $x^{\text{WB}*} = [A_m, \delta, \Delta p] = [10, 0.1, 1]$ and $x^{\text{BB}*} = [\text{FFV}, d_{\text{spacing}}] = [0.3, 5.13]$, yielding $J^* = 10.997$. The optimizer drives all white-box variables to their lower bounds—minimizing membrane area (capital cost) and pressure difference (operating cost) while maximizing flux through minimum thickness—and selects maximum FFV for permeability while tuning chain spacing to optimize the selectivity-permeability trade-off near the Gaussian peak at $d_{\text{spacing}} \approx 5 \text{ \AA}$.

A.13 Williams-Otto Process

We consider the Williams-Otto process [92, 93], a benchmark chemical process consisting of a continuously stirred tank reactor (CSTR) with three reactions:



where P is the desired product and G is waste. We formulate this as a multi-scale grey-box problem where catalyst properties (black-box from DFT calculations) affect kinetic selectivity, while the reactor mass balances at steady state are known white-box equations. The problem can be formulated as:

$$\begin{aligned} & \max_{T, F_{fB}, \mu, \varepsilon, \sigma_c} F_{pP} \\ \text{s.t. } & k_i(T) = \frac{a_i}{\rho} \exp\left(-\frac{b_i}{T}\right), \quad i \in \{1, 2, 3\}, \\ & r_1 = y_1 \cdot k_1(T) \cdot X_A X_B, \quad r_2 = y_2 \cdot k_2(T) \cdot X_B X_C, \quad r_3 = k_3(T) \cdot X_C X_P, \\ & 0 = F_{fA} - \mu X_A - r_1 \cdot m, \\ & 0 = F_{fB} - \mu X_B - r_1 \cdot m - r_2 \cdot m, \\ & 0 = -\mu X_C + 2r_1 \cdot m - 2r_2 \cdot m - r_3 \cdot m, \\ & 0 = -\mu X_E + 2r_2 \cdot m, \\ & 0 = -\mu X_P + r_2 \cdot m - 0.5r_3 \cdot m, \\ & 0 = -\mu X_G + 1.5r_3 \cdot m, \\ & F_{pP} = \mu X_P - 0.1\mu X_E, \quad F_{wG} = \mu X_G, \\ & g_1(x, y) := F_{wG} - 1.0 \leq 0, \\ & g_2(x, y) := 0.5 - X_A - X_B - X_C - X_E - X_P - X_G \leq 0, \\ & y_1 = f_1^{\text{BB}}(\varepsilon, \sigma_c) := \exp\left(-\frac{(\varepsilon - 15)^2}{50}\right) \cdot (1 + 0.1(\sigma_c - 5)), \\ & y_2 = f_2^{\text{BB}}(\varepsilon, \sigma_c) := 1.5 \cdot \exp\left(-\frac{(\varepsilon - 12)^2}{40}\right) \cdot \exp\left(-\frac{(\sigma_c - 6)^2}{8}\right), \\ & T \in [400, 700] \text{ }^\circ\text{R}, \quad F_{fB} \in [5, 50] \text{ klb/h}, \quad \mu \in [50, 200] \text{ klb/h}, \\ & \varepsilon \in [5, 25] \text{ kJ/mol}, \quad \sigma_c \in [3, 10] \text{ \AA}, \\ & x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [T, F_{fB}, \mu], \quad x^{\text{BB}} = [\varepsilon, \sigma_c], \end{aligned} \tag{18}$$

where $X_i = m_i/m$ are mass fractions and m is total reactor mass. [Refer to the parameters in Table 11.](#)

The black-box outputs y_1 and y_2 represent catalyst activity factors for reactions 1 and 2, modeled as volcano-type functions of catalyst surface energy ε and pore size σ_c . The constraint g_1 limits waste production ($F_{wG} \leq 1.0$ klb/h) and g_2 ensures mass balance closure ($\sum_i X_i \geq 0.5$). The global maximum net product flow is ~~5.460~~ 5.470 klb/h with ~~$x^{\text{WB}\star} = [581.6, 50.0, 50.0]^\top$~~ , ~~$x^{\text{BB}\star} = [12.61, 6.11]^\top$~~ $x^{\text{WB}\star} = [581.9, 50.0, 50.0]^\top$ $x^{\text{BB}\star} = [12.6, 6.11]^\top$.

Appendix B Acquisition Function Details

All BO experiments use Expected Improvement (EI) as the acquisition function. We also implemented four alternatives for completeness; the EI results are reported in the main text because EI is the most widely used baseline and enables direct comparison with prior work.

Table 11: Variables, physical meaning, bounds, and units (Williams-Otto Process)

Symbol	Description	Range / Value	Units
White-box decision variables			
T	Reactor temperature	[400, 700]	°R
F_{fB}	Feed flow rate of B	[5, 50]	klb/h
μ	Total outlet flow rate	[50, 200]	klb/h
Black-box decision variables			
ε	Catalyst surface energy	[5, 25]	kJ/mol
σ_c	Catalyst pore size	[3, 10]	Å
Black-box outputs			
y_1	Activity factor for reaction 1 ($A + B \rightarrow C$)	—	—
y_2	Activity factor for reaction 2 ($C + B \rightarrow P + E$)	—	—
Fixed parameters			
a_1	Pre-exponential factor (reaction 1)	5.9755×10^9	h^{-1}
a_2	Pre-exponential factor (reaction 2)	2.5962×10^{12}	h^{-1}
a_3	Pre-exponential factor (reaction 3)	9.6283×10^{15}	h^{-1}
b_1	Activation temperature (reaction 1)	12000	°R
b_2	Activation temperature (reaction 2)	15000	°R
b_3	Activation temperature (reaction 3)	20000	°R
ρ	Density	50	lb/ft ³
F_{fA}	Feed flow rate of A (fixed)	10	klb/h

B.1 Expected Improvement

For minimization, Expected Improvement is defined as

$$\alpha_{\text{EI}}(x) = (\hat{y} - \mu(x) - \xi) \Phi(Z) + \sigma(x) \phi(Z), \quad Z = \frac{\hat{y} - \mu(x) - \xi}{\sigma(x)}, \quad (19)$$

where $\mu(x)$ and $\sigma(x)$ are the GP posterior mean and standard deviation, $\hat{y}$ is the best observed objective value, Φ and ϕ are the standard normal CDF and PDF, and $\xi \geq 0$ is an exploration parameter. When $\sigma(x) = 0$, we set $\alpha_{\text{EI}}(x) = 0$. For maximization, the improvement direction is reversed: $Z = (\mu(x) - \hat{y} - \xi)/\sigma(x)$.

The exploration parameter ξ controls the trade-off between exploitation ($\xi = 0$, selecting points where the GP mean is low) and exploration ($\xi > 0$, favoring regions of high uncertainty even if the mean is not promising). We sweep $\xi \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$ in all experiments.

B.2 Probability of Improvement

Probability of Improvement measures the probability that a candidate point improves over the current best:

$$\alpha_{\text{PI}}(x) = \Phi(Z), \quad Z = \frac{\hat{y} - \mu(x) - \xi}{\sigma(x)}. \quad (20)$$

PI is a simpler alternative to EI that ignores the magnitude of improvement. It tends to exploit more aggressively than EI for the same ξ .

B.3 Lower Confidence Bound

The Lower Confidence Bound (LCB) acquisition function for minimization is

$$\alpha_{\text{LCB}}(x) = -(\mu(x) - \kappa \sigma(x)), \quad (21)$$

where $\kappa > 0$ controls exploration. We negate LCB so that all acquisition functions follow the convention that higher values are better (i.e., the next point is selected by $\arg \max_x \alpha(x)$). For maximization, the Upper Confidence Bound $\alpha_{\text{UCB}}(x) = \mu(x) + \kappa \sigma(x)$ is used. We set $\kappa = 2.0$ by default.

B.4 Modified Watson-Barnes 2

The modified Watson-Barnes 2 (mWB2) acquisition function was proposed in the COBALT framework [6]:

$$\alpha_{\text{mWB2}}(x) = s_n \cdot \alpha_{\text{EI}}(x) - \mu(x), \quad s_n = -3 \left(1 - \frac{n}{N}\right), \quad (22)$$

where n is the current iteration and N is the total budget. The dynamic scaling factor s_n starts large (emphasizing EI-driven exploration) and decreases toward zero (shifting weight to the GP mean for exploitation). For maximization, the sign of $\mu(x)$ is ~~flipped~~reversed.

B.5 Thompson Sampling

Thompson Sampling draws a sample function $\tilde{f}$ from the GP posterior and selects the point that optimizes the sample:

$$x_{\text{next}} = \arg \min_{x \in \mathcal{X}} \tilde{f}(x), \quad \tilde{f} \sim \mathcal{GP}(\mu, k). \quad (23)$$

In practice, we evaluate $\tilde{f}$ at a discrete set of candidate points by drawing from the multivariate normal $\mathcal{N}(\mu(X_{\text{grid}}), K(X_{\text{grid}}, X_{\text{grid}}))$. Exploration arises naturally from the randomness of the posterior draw rather than from an explicit exploration parameter.

B.6 Acquisition Optimization

For all acquisition functions, we maximize $\alpha(x)$ over a random grid of 1,000 candidate points drawn uniformly from the search domain (either $\mathcal{X}^{\text{BB}}$ for bilevel BO or $\mathcal{X}^{\text{WB}} \times \mathcal{X}^{\text{BB}}$ for black-box BO). This grid search approach avoids the gradient-based optimization of the acquisition function that can be problematic in low dimensions with multi-modal acquisition landscapes. The candidate with the highest acquisition value is selected as the next evaluation point.

Appendix C Gaussian Process Configuration

Both BO solvers use `scikit-learn`’s `GaussianProcessRegressor` with identical configuration. The kernel is a product of a constant kernel and an ARD RBF kernel, plus additive white noise:

$$k(x, x') = \sigma_f^2 \prod_{d=1}^D \exp\left(-\frac{(x_d - x'_d)^2}{2\ell_d^2}\right) + \sigma_n^2 \delta(x, x'), \quad (24)$$

where D is the input dimension (n_{BB} for bilevel BO, $n_{\text{WB}} + n_{\text{BB}}$ for black-box BO), σ_f^2 is the signal variance, ℓ_d is the length scale for dimension d (automatic relevance determination), and σ_n^2 is the noise variance.

Table 12 summarizes the hyperparameter bounds and optimization settings.

Table 12: GP hyperparameter configuration.

Parameter	Value / Bounds	Notes
Signal variance σ_f^2	$[10^{-10}, 10^3]$	Initial value 1.0
Length scales ℓ_d	$[10^{-10}, 10]$	One per input dimension (ARD)
Noise variance σ_n^2	$[10^{-10}, 10^{-1}]$	Initial value 10^{-5}
Regularization α	10^{-6}	Added to diagonal for numerical stability
Normalize y	True	Observations standardized before fitting
Optimizer restarts	5	L-BFGS-B on log marginal likelihood

The bilevel BO surrogate fits a GP over $n_{\text{BB}} = 1\text{--}2$ input dimensions, while the black-box BO surrogate fits over $n_{\text{WB}} + n_{\text{BB}} = 2\text{--}5$ dimensions. Since GP fitting scales as $O(N^3)$ in the number of observations N (which is

the same for both methods) and the per-prediction cost is $O(N^2D)$, the lower dimensionality of bilevel BO reduces prediction cost but does not change the dominant $O(N^3)$ training cost. In practice, with $N \leq 250$ and $D \leq 5$, GP fitting is fast for both methods (typically <0.1 seconds per iteration).

Appendix D ~~NLP-BH~~ Baseline Details

The black-box ~~NLP-BH~~ baseline solves the full problem (4) directly using ~~SciPy's `scipy.optimize.minimize` `basinhopping` with the SLSQP method.~~ Because SLSQP is a local solver and the problems are generally non-convex, we use a multi-start strategy with 20 random restarts drawn from a Latin hypercube over the full variable bounds ~~$\mathcal{X}^{\text{WB}} \times \mathcal{X}^{\text{BB}}$.~~

D.1 ~~Gradient~~ Computation

~~SLSQP requires gradient information. For the white-box variables x^{WB} , the objective J and constraints g are differentiable and `scipy` computes gradients via finite differences. For the black-box variables x^{BB} , the gradients of the composite function $J(x^{\text{WB}}, f^{\text{BB}}(x^{\text{BB}}))$ with respect to x^{BB} also require finite differences through f^{BB} . Each gradient evaluation thus requires $O(n_{\text{BB}})$ additional calls to f^{BB} . Over 20 multi-start runs, the total number of f^{BB} evaluations is typically 900–3space $[x^{\text{WB}}, x^{\text{BB}}]$ with an SLSQP local minimizer, a Metropolis temperature of $T = 100$, full-range stepsize (uniform perturbation clipped to variable bounds), and 1,000 , depending on the problem and the number of iterationseach SLSQP run takes to converge~~Basin-Hopping iterations.

D.1 Constraint Handling

~~SLSQP handles inequality constraints natively through its sequential quadratic programming formulation. However, convergence to feasible points is not guaranteed for non-convex constraint sets. We observed that on several constrained problems, some multi-start runs converge to constraint-violating local optima. The NLP baseline reports the best objective across all 20 starts without checking feasibility, which means its reported regret may be optimistic when the best NLP solution is slightly infeasible~~Constraints are handled by the ~~SLSQP local minimizer within Basin-Hopping, which enforces $g(x^{\text{WB}}, y, x^{\text{BB}}) \leq 0$ as inequality constraints at each local minimization. After optimization, feasibility is verified: solutions satisfying all constraints to within 10^{-6} are considered feasible. If no feasible solution is found, the solver returns the solution with minimum constraint violation.~~

In contrast, both BO methods track feasibility explicitly. For black-box BO, constraints are handled via a penalty of 10^6 added to the objective when any $g_i(x) > 0$. For bilevel BO, the inner ~~NLP-BH~~ solver checks constraint satisfaction and returns a penalty value when the inner problem is infeasible for a given y .

D.2 Sample Efficiency Comparison

Table 13 compares the number of black-box evaluations across solvers. The ~~NLP baseline uses an order of magnitude~~~~BH baseline evaluates f^{BB} at every~~

local minimization within Basin-Hopping, resulting in substantially more f^{BB} evaluations than either BO method, making it impractical when each evaluation is truly expensive (e.g., DFT calculations taking hours).

Table 13: Black-box evaluations (f^{BB} calls) per solver.

Solver	f^{BB} evaluations	Notes
Black-box BH	iteration-dependent	Multiple f^{BB} calls per BH iteration
Black-box BO	$n_{\text{init}} + 200$	One f^{BB} per iteration
Bilevel BO (SLSQP)	$n_{\text{init}} + 200$	One f^{BB} per iteration
Bilevel BO (BH)	$n_{\text{init}} + 200$	One f^{BB} per iteration

Both All three BO methods call f^{BB} exactly once per iteration, so the total sample budget is $n_{\text{init}} + N$ where $N = 200$ is the number of BO iterations. With $n_{\text{init}} = 5$ (the default smallest setting), this gives 205 evaluations. Bilevel BO additionally solves an inner NLP. Both bilevel BO variants additionally solve an inner optimization problem at each iteration, but this uses only the white-box model f^{WB} (which is cheap) and does not require additional calls to f^{BB} .

D.3 Failure Modes

The NLP baseline fails in two distinct ways. First, on multimodal problems (Rastrigin), gradient-based optimization converges to local minima. The 20-BH baseline fails on Rastrigin, where multimodality in the full combined space challenges even global solvers: the BH baseline achieves a final regret of 3.38, worse than all BO methods. It also performs poorly on Distillation, where the tight constraint set and complex objective landscape limit convergence. On all other problems, the BH baseline approaches the global optimum but requires substantially more f^{BB} evaluations than the BO methods.

Appendix E Inner-Loop and Full-Space Solver Benchmarks

Basin-Hopping (BH) was selected as the unified global solver for both the inner-loop white-box subproblem and the full-space NLP baseline. To justify this choice, we benchmarked five solvers across all 13 problems in both contexts: multi-start restarts mitigate this but do not eliminate it: the NLP achieves a final regret. SLSQP with 50 Latin hypercube restarts (SLSQP-50), differential evolution (DE) [94], simplicial homology global optimization (SHGO) [95], dual annealing (DA) [96], and Basin-Hopping with 100 iterations (BH). We additionally ran a high-iteration Basin-Hopping variant with 1,000 iterations (BH-1000) to assess convergence with extended computation.

E.1 Inner-Loop Solver Comparison

Table 14 reports regret (distance from verified global optimum) for each solver on the inner white-box subproblem, where x^{BB} is fixed and only x^{WB} is optimized. All five solvers have access to analytical gradients of J and g with respect to x^{WB} .

Table 14: Inner-loop solver regret. A dash (—) indicates the solver returned an infeasible solution. All feasible regret values are near machine precision ($< 10^{-7}$) except for DA, which fails to converge on six problems.

Problem	SLSQP-50	DE	SHGO	DA	BH	BH-1000
Small-Feas.-Region-1	4.4e-16	—	—	0	4.4e-16	4.4e-16
Small-Feas.-Region-2	0	4.4e-16	0	0	0	0
Rastrigin	1.8e-05	0	0	3.6e-05	1.4e-04	0
Toy-Hydrology	2.7e-08	2.7e-08	2.7e-08	—	2.7e-08	2.7e-08
Rosen-Suzuki	1.6e-14	1.2e-14	0	—	4.0e-15	3.9e-15
CSTR	0	8.1e-14	0	4.5e-14	1.5e-16	0
Heat-Exchanger	0	6.7e-14	0	0	0	0
PSA	0	3.0e-14	1.7e-16	4.3e-09	0	0
Batch-Reactor	2.0e-12	2.1e-12	2.0e-12	—	1.9e-12	2.0e-12
Distillation	8.2e-09	8.4e-08	—	—	8.5e-08	8.4e-08
Evaporator	1.0e-11	1.0e-11	1.0e-11	—	1.0e-11	1.0e-11
Membrane	0	6.7e-14	0	0	0	0
Williams-Otto	2.7e-11	5.0e-13	4.3e-13	—	4.3e-13	4.3e-13
Feasible	13/13	12/13	11/13	7/13	13/13	13/13

SLSQP-50 achieves the lowest regret on 7 of 13 problems, leveraging gradient information most directly. However, it is inapplicable to the derivative-free full-space problem (Section E.2). SHGO and DA suffer from feasibility failures: SHGO returns infeasible solutions on 2 problems (Small-Feasible-Region-1 and Distillation), while DA fails on 6 problems (Toy-Hydrology, worse than both BO methods. Second, on problems with narrow feasible regions or nonlinear black-box coupling (Small-Feasible-Region-2) Rosen-Suzuki, finite-difference gradients through f^{BB} are inaccurate near constraint boundaries. Batch-Reactor, Distillation, Evaporator, and Williams-Otto). DE is reliable (12/13 feasible) but misses feasibility on one problem. BH and BH-1000 both achieve feasibility on all 13 problems. The regret differences among feasible solvers are negligible in practice—on the order of 10^{-14} to 10^{-8} —well below any meaningful engineering threshold.

Table 15 reports the corresponding wall times. SLSQP-50 and DA are the fastest (median < 0.1 s), while SHGO is the slowest (median 3.1 s), with several problems exceeding 20 s. BH is fast at 100 iterations (median 0.20 s) and remains practical at 1,000 iterations (median 0.45 s), except for Williams-Otto (30.9 s) and Small-Feasible-Region-1 (27.0 s).

E.2 Full-Space Solver Comparison

Table 16 reports regret when each solver is applied to the full joint problem over $[x^{\text{WB}}, x^{\text{BB}}]$. SLSQP-50 is inapplicable in this context because the black-box function f^{BB} does not provide analytical gradients.

The NLP reports a regret full-space results exhibit sharper solver differentiation. SHGO fails on 7 of 13 problems, with 5 hitting the 60-second timeout on problems with higher dimensionality or tight constraints (Rastrigin, CSTR, Heat-Exchanger, PSA, Distillation, Membrane). DA fails on 9 of 13 problems and returns large-regret solutions even when feasible (e.g., 3.9×10^6 on PSA). DE is fully reliable (13/13 feasible) with uniformly low regret, but is the slowest among the feasible solvers.

BH achieves feasibility on all 13 problems with near-zero regret on 12 of

Table 15: Inner-loop solver wall time (seconds).

Problem	SLSQP-50	DE	SHGO	DA	BH	BH-1000
Small-Feas.-Region-1	1.694	1.158	0.488	0.024	0.314	27.049
Small-Feas.-Region-2	0.008	0.015	0.977	0.025	0.097	0.167
Rastrigin	0.012	0.058	25.235	0.050	0.121	0.200
Toy-Hydrology	0.028	0.034	0.601	0.024	0.204	0.505
Rosen-Suzuki	0.040	0.196	0.628	0.046	0.208	0.930
CSTR	0.023	0.284	13.349	0.079	0.247	0.607
Heat-Exchanger	0.009	0.246	3.054	0.070	0.104	0.170
PSA	0.030	0.295	1.627	0.072	0.236	0.448
Batch-Reactor	0.067	0.523	3.965	0.075	0.180	1.093
Distillation	0.016	0.433	21.461	0.068	0.168	0.375
Evaporator	0.018	0.382	0.866	0.073	0.207	0.336
Membrane	0.017	0.444	22.310	0.070	0.124	0.275
Williams-Otto	1.749	6.279	4.502	1.088	3.099	30.904
Median	0.023	0.295	3.054	0.070	0.204	0.448

Table 16: Full-space solver regret. A dash (—) indicates the solver returned an infeasible solution or is inapplicable. BH (100 iterations) fails catastrophically on Rastrigin due to multimodality in the combined space, but BH-1000 recovers.

Problem	SLSQP-50	DE	SHGO	DA	BH	BH-1000
Small-Feas.-Region-1	—	7.3e-09	7.3e-09	—	7.3e-09	7.3e-09
Small-Feas.-Region-2	—	3.8e-15	2.2e-16	3.6e-13	6.7e-16	0
Rastrigin	—	0	—	7.5e-04	9.9e+09	0
Toy-Hydrology	—	2.7e-08	—	—	2.7e-08	2.7e-08
Rosen-Suzuki	—	1.0e-13	1.3e-12	—	3.1e-15	3.7e-15
CSTR	—	1.5e-13	—	2.0e-12	3.1e-16	0
Heat-Exchanger	—	1.4e-13	—	—	0	0
PSA	—	1.7e-13	—	—	0	0
Batch-Reactor	—	2.2e-12	1.7e-12	—	1.9e-12	1.9e-12
Distillation	—	8.4e-08	—	—	8.6e-08	8.5e-08
Evaporator	—	1.0e-11	9.8e-12	—	9.8e-12	9.8e-12
Membrane	—	2.5e-13	—	1.8e-10	0	0
Williams-Otto	—	5.9e-13	3.1e-13	—	4.4e-13	4.4e-13
Feasible	0/13	13/13	6/13	4/13	13/13	13/13

them. The exception is Rastrigin, where BH at 100 iterations converges to a poor local optimum with a regret of 9.9×10^9 . This failure is resolved by increasing to 1,000 iterations (BH-1000), which achieves zero regret. This is the only problem where BH requires more than 100 iterations to find the global optimum in the full space.

E.3 Solver Selection Rationale

Table 17 summarizes the aggregate performance. BH is the only solver that achieves feasibility on all 13 problems in both contexts while also being applicable to derivative-free problems. Although SLSQP-50 wins on 7 of 13 inner-loop problems, it cannot be used for the full-space baseline. DE is the next most reliable solver (13/13 feasible in both contexts) but is slower (median 0.97 s vs. 0.39 s for BH in the full space). SHGO and DA are unsuitable as general-purpose solvers due to frequent feasibility failures.

Table 17: Aggregate solver statistics across all 13 benchmark problems. Feasibility counts and median regret are computed over feasible solutions only. BH provides the best combination of universal feasibility, near-exact optimality, and broad applicability.

Solver	Inner-loop		Full-space		Grad.-free
	Feasible	Med. regret	Feasible	Med. regret	
SLSQP-50	13/13	1.6e-14	—	—	No
DE	12/13	7.4e-14	13/13	2.5e-13	Yes
SHGO	11/13	0	6/13	1.5e-12	Yes
DA	7/13	0	4/13	9.2e-11	Yes
BH	13/13	4.0e-15	13/13	4.4e-13	Yes
BH-1000	13/13	4.4e-16	13/13	3.7e-15	Yes

The regret gap between BH and the best solver on any given problem is negligible—typically 10^{-14} to 10^{-8} —confirming that the choice of BH does not compromise solution quality. Because BH wraps an SLSQP local minimizer inside a global exploration loop, it inherits gradient-based precision on the inner problem while providing the global search capability needed for the full-space baseline. This combination of universal feasibility, near-exact optimality, and applicability to both derivative-available and derivative-free contexts makes BH the most defensible single-solver choice for the benchmark suite.

Appendix F Global Optimum Verification

Each problem’s global optimum J^* was verified through a multi-stage process designed to provide high confidence that the reported optima are correct.

F.1 Verification Procedure

We used ~~three complementary approaches~~ two complementary approaches, each repeated across 10 independent random seeds. First, we ran ~~differential evolution (DE)~~ Basin-Hopping (BH) over the full variable space $[x^{\text{WB}}, x^{\text{BB}}]$ with ~~a population of 100 and 1~~ an SLSQP local minimizer, a Metropolis temperature of $T = 100$.

full-range stepsize, and 2,000 generations. Second, we ran bilevel DE—differential evolution over x^{BB} with the inner NLP solved exactly for each candidate—to exploit the separable structure during verification. Third, BH iterations. Constraints were enforced by the SLSQP local minimizer. Second, we ran multi-start NLP with 1,000 random restarts SLSQP with 500 Latin hypercube starting points per seed.

For each problem, we compared the best solutions from all three approaches. When all three methods agreed to within 10^{-4} , we accepted the optimum. When they disagreed, we investigated manually by examining the objective landscape and constraint activity at the candidate optima.

F.2 Feasibility Checks

feasible solutions across all 20 runs (10 BH + 10 multi-start NLP). All 13 problems achieved agreement to within 2×10^{-4} relative error, confirming the reported optima. For constrained problems, we verified that the reported optimal solution satisfies all inequality constraints $g_i(x^{\text{WB}\star}, y^\star) \leq 0$ to within a tolerance of 10^{-6} .

We also checked that no nearby feasible point achieves a better objective, by perturbing the optimal solution along the active constraint boundaries and confirming that the objective worsens.

F.2 Verified Optima

Table 18 reports the verified global optimum for each problem, along with the optimal decision variables.

Table 18: Verified global optima for all 13 benchmark problems. Optima verified via Basin-Hopping (SLSQP local minimizer, $T = 100$, 2000 iterations, 10 seeds) and multi-start SLSQP (500 Latin hypercube starts, 10 seeds).

Problem	$J^\star$	$x^{\text{WB}\star}$	$x^{\text{BB}\star}$
Small-Feasible-Region 1	0.2532	[4.712]	[1.253]
Small-Feasible-Region 2	−2.000	[4.712]	[5.500]
Rastrigin	0.0000	[0, 0]	[0]
Toy-Hydrology	0.5998	[0.405]	[0.195]
Rosen-Suzuki	−44.00	[0, 1]	[2, −1]
CSTR	92.32%	[647.3, 20.0, 5.0]	[−1.047, 0.200]
Heat-Exchanger	1000.0	[1.0, 0.1, 19.3]	[50.0, 0.0]
PSA	−0.6433	[3.97, 0.1, 10.0]	[6.04, 19.6]
Batch-Reactor	−1.749	[500, 4.39, 5.0]	[−1.01, 2.00]
Distillation	11,758	[1.0, 11.0, 0.5]	[19.8, 9.99]
Evaporator	−1.959	[4.09, 19.1, 5.0]	[120.5, 2.0]
Membrane	11.00	[10.0, 0.1, 1.0]	[0.30, 5.13]
Williams-Otto	5.470	[581.9, 50.0, 50.0]	[12.6, 6.11]

Appendix G Proof of Proposition 1

Proof. We show that $\inf(\text{bilevel}) = \inf(\text{original})$ by constructing mappings in both directions.

(i) *Any feasible point of (4) maps to a feasible outer-level point with equal or worse objective.* Let $(x^{\text{WB}}, x^{\text{BB}})$ be feasible for (4) with $y = f^{\text{BB}}(x^{\text{BB}})$. By Assumption 1(ii), the inner problem (5c)–(5f) at this x^{BB} and y has a global solution $x^{\text{WB}\star}$ satisfying $J(x^{\text{WB}\star}, y, x^{\text{BB}}) \leq J(x^{\text{WB}}, y, x^{\text{BB}})$. Thus x^{BB} is outer-feasible with objective value no worse than the original point.

(ii) *Any outer-level feasible point maps back to a feasible point of (4) with equal objective.* Let $\tilde{x}^{\text{BB}}$ be outer-feasible with $\tilde{y} = f^{\text{BB}}(\tilde{x}^{\text{BB}})$ and inner solution $\tilde{x}^{\text{WB}\star}$. By construction, $\tilde{x}^{\text{WB}\star}$ satisfies $f^{\text{WB}}(\tilde{x}^{\text{WB}\star}, \tilde{y}) = 0$, $g(\tilde{x}^{\text{WB}\star}, \tilde{y}, \tilde{x}^{\text{BB}}) \leq 0$, and $\tilde{x}^{\text{WB}\star} \in \mathcal{X}^{\text{WB}}$. Together with $\tilde{x}^{\text{BB}} \in \mathcal{X}^{\text{BB}}$, the pair $(\tilde{x}^{\text{WB}\star}, \tilde{x}^{\text{BB}})$ is feasible for (4) with objective $J(\tilde{x}^{\text{WB}\star}, \tilde{y}, \tilde{x}^{\text{BB}})$.

(iii) *Infeasibility.* If the inner [NLP-problem](#) is infeasible for some $\tilde{y} \in \text{range}(f^{\text{BB}})$ —i.e., no $x^{\text{WB}} \in \mathcal{X}^{\text{WB}}$ satisfies $f^{\text{WB}}(x^{\text{WB}}, \tilde{y}) = 0$ and $g(x^{\text{WB}}, \tilde{y}, \tilde{x}^{\text{BB}}) \leq 0$ —then no feasible point of (4) exists at this $\tilde{x}^{\text{BB}}$ either, so excluding it from the outer search does not lose optimality.

From (i) and (ii), the feasible sets project onto each other with preserved objectives, so $\inf(\text{bilevel}) = \inf(\text{original})$. [To see how a global optimum of \(4\) appears in \(5\): let \$\(x^{\text{WB}\star}, x^{\text{BB}\star}\)\$ be a global minimizer of \(4\) with \$y^\star = f^{\text{BB}}\(x^{\text{BB}\star}\)\$. By step \(i\), \$x^{\text{BB}\star}\$ is outer-feasible in \(5\), and the inner problem at \$x^{\text{BB}\star}\$ returns some \$\hat{x}^{\text{WB}}\$ with \$J\(\hat{x}^{\text{WB}}, y^\star, x^{\text{BB}\star}\) \leq J\(x^{\text{WB}\star}, y^\star, x^{\text{BB}\star}\)\$. Since \$\(x^{\text{WB}\star}, x^{\text{BB}\star}\)\$ is already globally optimal for \(4\), the inequality must hold with equality: \$\hat{x}^{\text{WB}} = x^{\text{WB}\star}\$ \(up to alternative optima with the same objective value\). Thus the bilevel formulation recovers both the optimal \$x^{\text{BB}\star}\$ at the outer level and the optimal \$x^{\text{WB}\star}\$ at the inner level.](#)

Assumption 1(ii) is used in step (i) to ensure the inner global optimum is attained, and in step (ii) to ensure the inner solution is globally optimal (not merely a local minimum). The surrogate in Algorithm 1 models [\$x^{\text{BB}} \mapsto J\(x^{\text{WB}\star}\(x^{\text{BB}}\), f^{\text{BB}}\(x^{\text{BB}}\), x^{\text{BB}}\)J\(x^{\text{WB}\star}\(x^{\text{BB}}\), f^{\text{BB}}\(x^{\text{BB}}\), x^{\text{BB}}\) : \mathbb{R}^{n_{\text{BB}}} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{\text{WB}}} \mapsto \mathbb{R}\$](#) , which is a function of n_{BB} variables. $\square$

Appendix H Implementation Details

Infeasibility handling. If the inner [NLP-problem](#) is infeasible for some y —because the black-box outputs lead to a white-box problem with no feasible solution—we return a large penalty value. The acquisition function naturally steers away from these regions without requiring explicit constraint modeling at the outer level.

Modularity. The inner [NLP-solver-optimizer](#) can be swapped ([Basin-Hopping](#), SLSQP, IPOPT, BARON) without changing the outer BO framework. Similarly, any acquisition function (EI, PI, LCB) can be used in the outer loop.

Table 19: Median iterations to reach 1% of initial regret for Bi-BO (SLSQP) ($n_{\text{init}} = 50$, best ξ , 10 repetitions). “>200” means the target was not reached within 200 iterations. Speedup values marked with $\geq$ are lower bounds (BB-BO did not converge).

Problem	BB-BO	Bi-BO (SL)	Speedup
Small-Feasible-Region	>200	>200	—
Small-Feasible-Region-2	>200	50	$\geq 4.0\times$
Rastrigin	>200	55	$\geq 3.6\times$
Toy-Hydrology	>200	59	$\geq 3.4\times$
Rosen-Suzuki	>200	134	$\geq 1.5\times$
CSTR	87	50	$1.8\times$
Heat-Exchanger	151	7	$20.1\times$
PSA	>200	134	$\geq 1.5\times$
Batch-Reactor	>200	>200	—
Distillation	>200	50	$\geq 4.0\times$
Evaporator	>200	>200	—
Membrane	>200	>200	—
Williams-Otto	>200	50	$\geq 4.0\times$

Table 20: Post-initialization BO iterations required for Bi-BO (SLSQP) to reach BB-BO’s median final regret, and vice versa ($n_{\text{init}} = 50$, 200-iteration BO budget, 10 repetitions, best ξ per problem per method). An iteration count of 0 means the target is reached at the end of the initial sampling phase. “>200” indicates the target is never reached within the budget in the majority of repetitions. Bi-BO’s NLP-assisted initial samples already dominate BB-BO’s full-budget final on 11 of 13 problems.

Problem	BB-BO final	Bi-BO final	Bi-BO iters $\rightarrow$ BB-BO	Iters saved	BB-BO
Small-Feasible-Region	4.77×10^{-3}	4.36×10^{-4}	54	+146	
Small-Feasible-Region-2	2.43×10^{-2}	3.09×10^{-6}	1	+199	
Rastrigin	1.85	2.65×10^{-3}	0	+200	
Toy-Hydrology	1.14×10^{-2}	1.05×10^{-6}	0	+200	
Rosen-Suzuki	4.88	2.43×10^{-2}	0	+200	
CSTR	3.38×10^{-1}	1.87×10^{-3}	1	+199	
Heat-Exchanger	3.71×10^1	3.64×10^{-7}	0	+200	
PSA	1.04×10^{-1}	2.99×10^{-3}	0	+200	
Batch-Reactor	2.89×10^{-1}	4.63×10^{-4}	0	+200	
Distillation	3.62×10^3	3.37×10^{-3}	0	+200	
Evaporator	3.84×10^{-1}	1.45×10^{-2}	1	+199	
Membrane	1.24	4.28×10^{-5}	0	+200	
Williams-Otto	8.25×10^{-1}	7.94×10^{-6}	0	+200	

Appendix I Convergence Speed Details

Appendix J Hyperparameter Sensitivity Details

We sweep over two key BO hyperparameters: the number of initial samples $n_{\text{init}} \in \{1, 5, 20, 50\}$ drawn from a Latin hypercube design, and the EI exploration parameter $\xi \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$. Each configuration is repeated 10 times with independent seeds 42–51 to quantify variability. Every BO run uses 200 iterations after initialization. In total, the experiment comprises ~~78,410–450~~ independent optimization runs: ~~$13 \times 4 \times 7 \times 10 \times 2 = 7,280$~~ ~~$13 \times 3 \times 7 \times 10 \times 3 = 8,190$~~ BO runs (~~one~~ black-box and ~~bilevel~~ two bilevel variants) plus $13 \times 10 = 130$ NLP runs (~~one and~~ $13 \times 10 = 130$ full-space BH runs (one each per problem-seed combination, since NLP and BH results do not depend on n_{init} or ξ).

Table 21: Bi-BO (SLSQP) mean final regret across n_{init} values (best ξ per problem per n_{init} , 10 repetitions).

Problem	$n_{\text{init}} = 5$	$n_{\text{init}} = 25$	$n_{\text{init}} = 50$
Small-Feasible-Region	0.0014	0.0011	0.0004
Small-Feasible-Region-2	0.0000	0.0000	0.0000
Rastrigin	0.0077	0.0063	0.0026
Toy-Hydrology	0.0000	0.0000	0.0000
Rosen-Suzuki	0.0078	0.0607	0.0243
CSTR	0.0015	0.0019	0.0019
Heat-Exchanger	0.0000	0.0000	0.0000
PSA	0.0013	0.0022	0.0030
Batch-Reactor	0.0003	0.0003	0.0005
Distillation	0.0081	0.0056	0.0034
Evaporator	0.0015	0.0085	0.0145
Membrane	0.0000	0.0001	0.0000
Williams-Otto	0.0000	0.0000	0.0000

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Table 22: Best EI exploration parameter ξ per problem for Bi-BO (SLSQP), selected by lowest mean final regret across all n_{init} values. These are the ξ values used to report results in Table 3.

Problem	best ξ
Small-Feasible-Region	0.01
Small-Feasible-Region-2	0.001
Rastrigin	1.0
Toy-Hydrology	0.001
Rosen-Suzuki	0.05
CSTR	0.001
Heat-Exchanger	0.001
PSA	0.001
Batch-Reactor	0.001
Distillation	0.001
Evaporator	0.5
Membrane	1.0
Williams-Otto	0.001

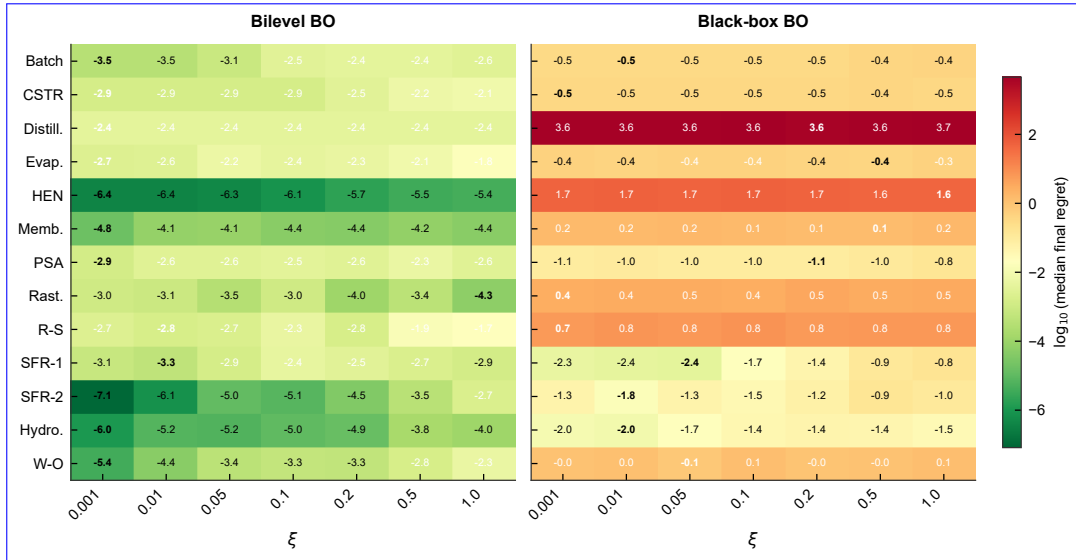


Figure 9: Sensitivity to the EI exploration parameter ξ . Left: bilevel BO; right: black-box BO. Rows are problems, columns are ξ values, aggregated over all n_{init} . Cell values show $\log_{10}(\text{median final regret})$; bold indicates best ξ per problem. No single ξ is universally optimal, but bilevel BO outperforms BB-BO regardless of ξ .

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